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OPTICAL ROUTING DEVICE FOR ULTRA-FLEXIBLE AND ULTRA-RELIABLE LASER BEAM BASED WIRELESS COMMUNICATION

Abstract

A first optical routing device that includes a mounting component, a memory configured to store optical path options, where the optical path options are different fall-back laser-link options available to the first optical routing device to establish a laser beam connectivity, and an optical routing component attached to the mounting component. The mounting component includes a rechargeable battery to power operations of the first optical routing device and a processor that communicates over-the-air with a master communication device or one or more service communication devices via RF supervisory links, receives an instruction via the one or more RF supervisory links to control a movement of the mounting component along with the optical routing component, and performs a range measurement to compute a distance between the first optical routing device and an optical node.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS/INCORPORATION BY REFERENCE
[0001] This patent application makes reference to, claims priority to, claims the benefit of, and is a Continuation application of U.S. patent application Ser. No. 18/746,965, filed on Jun. 18, 2024, which is a Continuation application of U.S. Pat. No. 12,114,113 issued on Oct. 8, 2024, which is a continuation application of U.S. Pat. No. 11,902,719 issued on Feb. 13, 2024, which is a continuation application of U.S. Pat. No. 11,800,261 issued on Oct. 24, 2023. Each of the above reference applications is hereby incorporated herein by reference in its entirety.

FIELD OF TECHNOLOGY

[0002] Certain embodiments of the disclosure relate to a wireless communication system. More specifically, certain embodiments of the disclosure relate to an optical routing device for ultra-flexible and ultra-reliable laser beam based wireless communication, for example, in an indoor area.

BACKGROUND

[0003] Conventional wireless communication devices, such as a wireless access point (WAP) or wireless routers, are often used in an indoor area to extend the wireless coverage of an existing Wi-Fi® signal to access Internet and to increase the numbers of end devices (users) that are capable to use Wi-Fi® may connect to the WAP. However, Wi-Fi® signals by virtue of the limitation of the Wi-Fi® communication protocol have a defined range beyond which the connectivity is lost. Thus, a large number of WAPs or range extenders are used in the indoor area (e.g., an enterprise building, a home) if wireless coverage for Wi-Fi® signals are to be extended. Moreover, under ideal conditions, typically 2.4 GHz Wi-Fi® supports up to 450 Mbps or 600 Mbps, and 5 GHz Wi-Fi® supports up to 1300 Mbps. Recently, progress has been made in developing Wi-Fi® communication in 6 GHz band, for example, for Wi-Fi® 6, 6E, Wi-Fi 7 etc., for increasing data throughput. However, the technical issue of limited coverage area and signal attenuation with increasing distance from a Wi-Fi® signal broadcast device remains. This is the reason why Ethernet cables are still widely used as primary connectors in an Ethernet Local Area Network (LAN) to connect different routers, switches, hub, and gateway devices in an enterprise to provide a reliable connectivity. Ethernet connects any computer or other electronic device to its network as long as the device has an Ethernet adapter or a network card. Since all the network devices need to

be connected to its individual power source, and cables need to run through walls, re-designing an enterprise LAN often becomes a herculean task. For instance, network maintenance may require changing or replacing cables and costly intermediate network devices that often causes damage to the walls and installation points for deployment or re-deployment.

[0004] Currently, certain attempts have been made to solve the technical issue of limited coverage area and signal attenuation with increasing distance from a Wi-Fi® signal broadcast device and to avoid the Ethernet cables. For example, certain conventional home mesh Wi-Fi® systems also exists that provide a comparatively wider coverage than the single Wi-Fi® signal broadcast device in an indoor area. Such conventional home mesh Wi-Fi® systems operate by broadcasting from the main router to several satellite routers, which in turn broadcast to its client devices. In this topology, the communication between the main router connected to the Internet and the satellite routers fed by the main router, is carried by the Backhaul. While the backhaul connections can be done with physical cables which offer wider bandwidth than a Wi-Fi® band, cable installation and reconfiguration are extremely costly and slow. Thus, there are many technical challenges associated with existing wireless communication systems commonly used in an indoor area. In a first example, there is the technical issue of how to provide full coverage with high signal-to-noise (SNR) in an indoor area with high-speed data connectivity to end-user devices without the need to deploy intermediate routers that process signals, for example, for signal amplification and routing. In another example, the reach and performance of existing systems (e.g., home mesh Wi-Fi® systems) are fundamentally limited by the bandwidth of a dedicated band, and by the number of intermediate routers (i.e., satellite routers) that the data must traverse in order to reach remote end-user devices. In a third example, currently, certain communication devices, such as Internet-of-Things (IoT) devices depend on high-speed Internet access to the cloud to send sensor data and receive instructions (e.g., artificial intelligence-based processing models) from cloud either directly or via a gateway device. The number of wireless sensors and IoT devices are rapidly increasing with the increase in smart homes, smart offices, enterprises, etc. Existing communication systems and Wi-Fi® standards are unable to handle such massive number of wireless sensors and IoT devices and their quality-of-service (QOS) requirements. In such cases, it is extremely difficult and technically challenging to support these end user devices, where latency and signal noise are other technical problem with existing communication systems and network architecture when operating in a sub 10 GHz frequency, and such latency increases when more wireless access points or relay nodes or routers are introduced in the network to extend the communication range. Furthermore, it is known that certain optical beam deflectors may be used to extend the range of an optical communication system. However, conventional optical beam deflectors are mainly designed and developed for space applications or other outdoor use cases, and inherently are very complex, costly, and not practically suited for end use, like in homes or general enterprises. Moreover, conventional optical beam deflectors are either bulky, very power intensive devices, or simply are not usable or do not work as required in a laser beam based wireless communication in indoor scenarios.

[0005] Further limitations and disadvantages of conventional and traditional approaches will become apparent to one of skill in the art, through comparison of such systems with some aspects of the present disclosure as set forth in the remainder of the present application with reference to the drawings.

BRIEF SUMMARY OF THE DISCLOSURE

[0006] An optical routing device for ultra-flexible and ultra-reliable laser beam based wireless communication in an indoor area, substantially as shown in and/or described in connection with at least one of the figures, as set forth more completely in the claims.

[0007] These and other advantages, aspects and novel features of the present disclosure, as well as details of an illustrated embodiment thereof, will be more fully understood from the following description and drawings.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0008] FIG. 1 is a diagram illustrating an exemplary communication system with optical routing devices, in accordance with an exemplary embodiment of the disclosure.

[0009] FIG. 2 is a block diagram that illustrates various components of an exemplary master communication device of a communication system, in accordance with an exemplary embodiment of the disclosure.

[0010] FIG. 3A is a block diagram that illustrates various components of an optical routing device of a communication system, in accordance with an exemplary embodiment of the disclosure.

[0011] FIG. 3B is a diagram illustrating an exemplary optical routing device of a communication system, in accordance with another exemplary embodiment of the disclosure.

[0012] FIG. 3C is a block diagram illustrating various components of an exemplary optical routing device of a communication system, in accordance with yet another exemplary embodiment of the disclosure.

[0013] FIG. 4 is a block diagram that illustrates various components of an exemplary service communication device of a communication system, in accordance with an exemplary embodiment of the disclosure.

[0014] FIG. 5 is a block diagram that illustrates various components of an exemplary cloud server of a communication system, in accordance with an exemplary embodiment of the disclosure.

[0015] FIG. 6A is a diagram illustrating an exemplary implementation of a communication system with optical routing devices for ultra-flexible and ultra-reliable laser beam based wireless communication in an indoor area, in accordance with an exemplary embodiment of the disclosure.

[0016] FIG. 6B is a diagram illustrating another exemplary implementation of a communication system with optical routing devices for ultra-flexible and ultra-reliable laser beam based wireless communication in an indoor area, in accordance with an exemplary embodiment of the disclosure.

[0017] FIGS. 7A, 7B, and 7C collectively, is a flowchart that illustrates an exemplary method for operating an optical routing device for ultra-flexible and ultra-reliable laser beam based wireless communication for indoor use, in accordance with an embodiment of the disclosure.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0018] Certain embodiments of the disclosure may be found in an optical routing device for ultra-flexible and ultra-reliable laser beam based wireless communication, for example, in an indoor area. The optical routing device is a new device that smartly employs one or more radio frequency (RF) supervisory links to communicate with a master communication device or one or more service communication devices for monitoring and control purpose while employs a laser beam deflection capability for actual laser beam based wireless communication in a multigigabit data rate to form a laser beam based wireless network in an indoor area. The optical routing device ensures an ultra-flexible and ultra-reliable laser beam based wireless communication with significantly lower power consumption, which can be powered even by an internal power source, such as an inbuilt battery. For example, with just one external power source, such as the master communication device plugged into an external power source (e.g., an AC current), all other optical routing devices and even some service communication device may be charged remotely via optical wireless power transmission by the master communication device **102**, thereby simplifying the deployment.

Furthermore, beneficially, the optical routing device comprises an optical routing component that includes one or more laser beam handling regions (which performs laser beam deflection over-the-air and in some cases free-space laser beam splitting or combining) configured to handle a plurality of laser beams concurrently in which a first laser beam is deflected via a first region of the one or more laser beam handling regions for downstream data communication in a downstream path and a second laser beam is deflected via a second region of the one or more laser beam handling regions

for upstream data communication in an upstream path. This not only improves network redundancy and network scalability but also enables almost near zero latency communication and an always-connected experience. The network redundancy refers to a mechanism or a way to provide multiple paths for traffic, so that data can keep flowing even in the event of a failure, and thus high network redundancy means high reliability. The term network scalability refers to an ability of a communication network to handle increasing workloads in a cost-effective and sustainable way, for example, by expanding the network's bandwidth capacity and supporting its physical expansion to new physical areas. The disclosed optical routing device is a new passive optical node that creates a smart laser-based wireless network in an indoor area with a wireless optical backhaul that ensures full coverage with high signal-to-noise (SNR) in different physical spaces including all nooks and corners of the indoor area with significantly improved high-speed data connectivity and data security to end-user devices without the need to deploy costly and bandwidth limited Wi-Fi® intermediate routers that process signals, for example, for signal amplification. The optical routing device which acts as a passive optical node of the communication system of the present disclosure further improves data transfer rates between at least two communication devices as compared to existing wireless systems (e.g., conventional wireless local area networks).

[0019] In the following description, reference is made to the accompanying drawings, which form a part hereof, and in which is shown, by way of illustration, various embodiments of the present disclosure.

[0020] FIG. 1 is a diagram illustrating an exemplary communication system with optical routing devices, in accordance with an exemplary embodiment of the disclosure. With reference to FIG. 1, there is shown a communication system **100** that may include a master communication device **102**, a plurality of optical routing devices **104**, one or more service communication devices **106**, a plurality of end-user devices **108** in a defined indoor area **112**. The plurality of end-user devices **108** may include one or more user equipment (UEs) **108A** and **108B**, and an image-capture device **110**. There is further shown a data source **114** connected to the master communication device **102**. In an implementation, the communication system **100** may include a cloud server **116** that is communicatively coupled to the master communication device **102** directly or via the data source **114**. There is further shown a first communication network **118** (e.g., Internet) and a second communication network **120** (i.e., a laser beam based wireless communication network). In an implementation, each optical node (i.e., each of the plurality of optical routing devices **104**, the master communication device **102**, and each of the one or more service communication device **106**) may include one or more sensors **122**.

[0021] Each of the master communication device **102** and the one or more service communication devices **106** may be a wireless access point or a wireless router, a home gateway device, a fixed wireless access (FWA) device, or a network controller.

[0022] Each of the plurality of optical routing devices **104** may be mounted on a ceiling surface or a wall surface. Beneficially, each of the plurality of optical routing devices **104** may have at least two distinct regions (e.g., deflection regions or laser beam handling regions) to handle a pair of laser beams concurrently in which a first laser beam deflected is used for downstream data communication in the downstream path and the second laser beam that is deflected is used for upstream communication in an upstream path. The downstream path refers to communication from the master communication device **102** towards the one or more service communication devices **106**. The upstream path refers to communication from the end-user devices to the one or more service communication devices **106** and further to the master communication device **102**. The master communication device **102** may be a modified and intelligent wireless access point that may use an existing Wi-Fi® signal in the RF medium to control one or more of the plurality of optical routing devices **104** whereas may employ laser beams to form the second communication network **120** (i.e., the laser beam based communication network) to wirelessly communicate data in a cascaded structure or a mesh structure to reach to the one or more service communication devices

106 via the plurality of optical routing devices **104**. Each of the plurality of optical routing devices **104** may be a battery-operated device, which is easy to deploy in the defined indoor area **112**. Further, each of the plurality of optical routing devices **104** may not only be capable of performing the deflection function by deflecting one or more laser beams but also may be configured to perform passive power splitting or combining (e.g., using Dichroic mirrors), passive polarization splitting or combining, and/or passive laser beam wavelength splitting or combining (e.g., using wavelength division multiplexing (WDM)).

[0023] The plurality of end-user devices **108** may include user equipment (UEs), or other consumer electronic devices, such as the image-capture device **110**. Each of one or more UEs, such as the UEs **108A** and **108B**, may correspond to a telecommunication hardware used by an end-user to communicate. Alternatively stated, the UEs **108A** and **108B**, may refer to a combination of a mobile equipment and subscriber identity module (SIM). Examples of the one or more UEs may include, but are not limited to a smartphone, a virtual reality headset, an augment reality device, a wireless modem, a satellite television set-top box, a VoIP station, or any other customized hardware for telecommunication and Internet connectivity.

[0024] The defined indoor area **112** may be an indoor area of an enterprise, an apartment, a building, or other indoor area of a building.

[0025] The data source **114A** may be a medium through which a backhaul connection is provided to the master communication device **102**. Examples of the data source **114A** may be a fiber optic cable, a high-speed XG communication wireless link, where “XG” refers to 5G or 6G, or a laser-based beam from an external communication device, such as an XG-enabled small cell or an XG-enabled repeater device located in an outdoor area. The modem **114B** may be a gateway device or a Wi-Fi® modem that combines the functionality of a modem and a router for Internet connectivity.

[0026] The cloud server **116** includes suitable logic, circuitry, and interfaces that may be configured to communicate with the master communication device **102**. In an example, the cloud server **116** may be a remote management server that is managed by a third party different from the service providers associated different cellular or wireless carrier networks or Internet service providers. In another example, the cloud server **116** may be a remote management server or a data center that is managed by a third party, or jointly managed, or managed in coordination and association with one or more of the cellular or wireless carrier networks or Internet service providers. In an implementation, the cloud server **116** may be a master cloud server or a master machine that is a part of a data center that controls an array of other cloud servers communicatively coupled to it, for load balancing, running customized applications, and efficient data management.

[0027] The first communication network **118** may include a medium through which the cloud server **116** and the master communication device **102** may communicate with each other. The master communication device **102** may be configured to obtain a first signal (e.g., data signal) from the data source **114A** or the modem **114B** coupled to the first communication network **118**. The data source **114A** or the modem **114B** may be used to provide an Internet connection via the first communication network **118** from an Internet service provider or from a core network of a telecommunication network. Examples of the first communication network **118** may include, but are not limited to, at least one of a Transmission Control Protocol and Internet Protocol (TCP/IP), Hypertext Transfer Protocol (HTTP), the Internet, a cloud network, an optical fiber, a coaxial cable, or other backhaul connectivity medium.

[0028] The second communication network **120** refers to a laser beam-based wireless communication network formed by the master communication device **102** in association with the plurality of optical routing devices **104** and the one or more service communication devices **106**.

[0029] The one or more sensors **122** may be provided at each optical node (i.e., each of the plurality of optical routing devices **104**, the master communication device **102**, and each of the one or more service communication device **106**). In some cases, some sensors of the one or more sensors **122** may be deployed external to the optical node in the defined indoor area **112**. Examples

of the one or more sensors **122** may include, but are not limited to magnetic and gravity field sensors, accelerometers, one or more motorized altazimuth, one or more Micro Electronic Mechanical Systems (MEMS) motion sensor, gyroscopes, a gimbal, one or more image-capture devices, a geospatial sensor (e.g., a GPS sensor), and the like.

[0030] In accordance with an embodiment, the master communication device **102** may be provided at a first location in the defined indoor area **112**. The master communication device **102** may be communicatively coupled to the cloud server **116** via the first communication network **118**. In an implementation, the master communication device **102** may be connected to the data source **114A** or the modem **114B**. In another implementation, the master communication device **102** may be integrated with the modem **114B** (i.e., the functionalities of a modem, such as a Wi-Fi® modem, may be integrated with the master communication device **102**). A service communication device, such as the service communication device **106A**, may be provided at a second location in the defined indoor area **112**. In an implementation, the service communication device **106A** may be provided at the second location where the signal strength is below a threshold value, for example less than 20 or 30 decibels. In certain scenarios, there may be no-line-of-sight (NLOS) between the master communication device **102** and the service communication device **106A**. In some implementation, multiple service communication devices, such as the one or more service communication devices **106**, may be provided in the defined indoor area **112**, where the signal strength is below a threshold value, e.g., a low SNR value, such as less than 20 or 30 decibels. The threshold value may be a default threshold value pre-defined in the master communication device **102** or may be user-configurable value that may be defined by a user. For example, an application installed in a smartphone may be used to connect to the master communication device **102** and set or update the threshold value. Alternatively, the threshold value may be remotely configured at the master communication device **102** from the cloud server **116**.

[0031] The plurality of optical routing devices **104** may be distributed at a plurality of locations in the defined indoor area **112** different from the first location and the second location. Each of the plurality of optical routing devices **104** may also be referred to as passive optical beam handling devices. Advantageously, the plurality of optical routing devices **104** may be battery-operated optical nodes. In an implementation, one or more of the plurality of optical routing devices **104** may be placed on or integrated with an electromechanical turret (also referred to as “eyeball”). The communication system **100** integrates optical wireless links that are reconfigurable via the battery-operated optical routing devices, such as the plurality of optical routing devices **104**, which routes data signals via laser beams (i.e., used for the optical wireless links) to different nodes, such as the one or more service communication devices **106**, according to instructions provided by the master communication device **102** or the cloud server **116**.

[0032] In an implementation, the master communication device **102** may be configured to indicate where to install the plurality of optical routing devices **104**. In order to indicate different installation positions of the plurality of optical routing devices **104**, the master communication device **102** may be configured to direct a pencil laser beam in a visible light spectrum to the optical routing device **104A** in the defined indoor area **112**. For example, the master communication device **102** may direct the pencil laser beam in the visible light spectrum pointing towards a point in a wall near the ceiling or a point in the ceiling in the defined indoor area **112**, where both points are in a line-of-sight of the master communication device **102**. After the optical routing device **104A** is installed at the selected point, the master communication device **102** may be configured to control a movement of the optical routing device **104A** and direct the pencil laser beam in the visible light spectrum towards the optical routing device **104A**, which further deflects the pencil laser beam to a plurality of points in the defined indoor area **112**. Based on at least where the service communication device, such as the service communication device **106A**, is to be placed, one point from the plurality of points is selected to install the optical routing device **104B** of the plurality of optical routing devices **104** (e.g., the optical routing devices **104A**, **104B**, **104C**, . . . , **104N**).

Similarly, the third, fourth, and other optical routing devices may be installed directed by the master communication device **102**.

[0033] In another example, the master communication device **102** may be configured to acquire a visual data item, such as an image, a video, or a three-dimensional (3D) map of the defined indoor area **112**. Such visual data item may be captured from the application installed in the smartphone held by a user, from any image-capture device, such as the image-capture device **110**. Such visual data item may be pre-captured or captured in real time and acquired by the master communication device **102**. Thereafter, the master communication device **102** may be further configured to process the visual data item to determine a plurality of different points in the defined indoor area **112** where the plurality of optical routing devices **104** may be placed. Such placement forms a cascaded or a mesh network of the plurality of optical routing devices **104** to cause the master communication device **102** to reach to the one or more service communication devices **106**. In another example, one or more 3D mapping cameras and digital twin technology can optionally be used to plan the location of the optical nodes, by creating an immersive 3D model of the operating environment. In yet another example, the plurality of optical routing devices **104** may be manually placed at the plurality of different locations in the defined indoor area **112** without assistance from the master communication device **102**.

[0034] Due to the line-of-sight (LOS) nature of laser beam transmission, misalignment of a laser transmitter and a laser receiver, known as a pointing error, may reduce data transmission fidelity or hinder such transmissions. In one aspect of the present disclosure, the installation of each optical node (i.e., each of the plurality of optical routing devices **104**, the master communication device **102**, and each of the one or more service communication device **106**) may be performed with proper alignment in a manner that is easy, robust, and can be done even by a non-technical person. The first step may be proper mounting of all the optical nodes (i.e., each of the plurality of optical routing devices **104**, the master communication device **102**, and each of the one or more service communication device **106**) on a hard surface such as a wall, ceiling, or pole. To this end, laser-levelling tools, may be used to project horizontal and vertical visible laser lines on such hard surfaces to guide the positioning of the optical nodes to ensure a rough or coarse initial alignment.

[0035] In an implementation, a coarse alignment may be executed using measurements of the Earth's magnetic field to work out the orientation (i.e., the heading) and the gravitational field to sense the tilt (i.e., the level) of the optical nodes including the laser transmitter, and the laser receiver. Each optical node (i.e., each of the plurality of optical routing devices **104**, the master communication device **102**, and each of the one or more service communication device **106**), may include one or more sensors **122**, such as magnetic and gravity field sensors, which may be used to measure the direction of the Magnetic North and a downward direction towards earth surface at the location of the one or more sensors **122**.

[0036] In an implementation, the one or more sensors **122** may include three magnetic field sensors, which may be provided at each optical node (i.e., each of the plurality of optical routing devices **104**, the master communication device **102**, and the one or more service communication device **106**). The three magnetic field sensors may be positioned perpendicular to each other, which may be used to find a local direction of the earth's Magnetic North. The magnetic field sensors may employ an optional declination correction to relate the magnetic north to the true north. Further, the one or more sensors **122** may include three accelerometers, which may sense gravity to provide tilt information (i.e., to evaluate which way is downwards). Using the orientation and the tilt (i.e., level) data, a motorized altazimuth (i.e., one of the one or more sensors **122**) may be used that may automatically point and align two optical nodes together to establish a wireless laser beam communication. In an example, the one or more sensors **122** may further include a Micro Electronic Mechanical Systems (MEMS) motion sensor, which may have a 3-axis accelerometer as well as a 3-axis gyroscope, may be used.

[0037] In another implementation, in order to achieve leveling, a gimbal similar to that used in

ships, to maintain horizontal levels for compasses, stoves, lights etc., may be employed. In yet another implementation, the one or more sensors **122** may include a miniature pendulum, which may be used to maintain a precise horizontal level with respect to the downward force on the pendulum. The plurality of the orientation, level and location data may be communicated to the master communication device **102** or the cloud server **116** (i.e., a network management software), for visualization, troubleshooting, optimization, and monitoring of the laser beam based wireless network, such as the second communication network **120**.

[0038] In accordance with an embodiment, after the coarse alignment, a fine alignment may be executed at each optical node. Based on the orientation and tilt sensor data, the master communication device **102** or the cloud server **116** may be further configured to perform the fine alignment at each optical node to align a laser beam transmission path at each optical node. The fine alignment may be performed by use of one or more beacon lasers (e.g., a pencil laser beam in the visible light spectrum), one or more image-capture devices, such as the image-capture device **110**, and a motorized altazimuth.

[0039] In an implementation, the master communication device **102** may be configured to direct a pencil laser beam in a visible light spectrum to the optical routing device **104A** in the defined indoor area **112** to align an optical path. The optical routing device **104A** may then deflect the pencil laser beam to other optical routing devices **104B**, **104C**, . . . , **104N**, in the defined indoor area **112**, to make them optically aligned to the laser beam transmission path. Similarly, each service communication device may be aligned accordingly. The optical alignment may be controlled by the master communication device **102** or the cloud server **116**. At each optical node, transmitted laser power may be monitored as well as the optical receiver RSSI. If laser power has not degraded but the optical receiver RSSI indicates a drop in power, then an automatic optical alignment routine may be executed. The optical receiver RSSI and transmit laser power may be monitored and reported back to the master communication device **102** or the cloud server **116** (hosting network management software) via a radio frequency (RF) wireless connectivity. Additionally, visible color LED's may be provided on each optical node to indicate a connection status. For example, when a laser beam link is operating properly, the LED may be green at such particular node. When power is below a defined level, the indicator may turn orange and when the laser link is interrupted, it may turn red. Further, when the communication system **100** may be executing an alignment procedure, the LED indicator may keep flashing.

[0040] In an implementation, the one or more sensors **122** at each optical node may also be equipped with a geospatial sensor (e.g., a GPS sensor) for the localization of the optical nodes. For identifying the locations of the optical nodes in GPS-denied environments, when GPS accuracy is insufficient, and for optimization of the laser beam, optical nodes may perform range measurement. The range measurements may provide the distance between two optical nodes (i.e., specifically, the laser receiver and laser transmitter) independent of the GPS sensor. The range measurements may be done by pulse or AC modulating the beacon laser and using time of flight measurements at the laser receiver using a common clock. The range measurement in combination with tunable lenses (in which focal length may be adjusted mechanically or electrically) may permit fine adjustments to the beam divergence for optimum data signal transmission across the laser beam based wireless network, such as the second communication network **120**.

[0041] Different embodiments and operations of optical routing devices of the plurality of optical routing devices **104** are described in detail, for example, in FIGS. 3A, 3B, and 3C. Advantageously, each of the plurality of optical routing devices **104** smartly employs one or more radio frequency (RF) supervisory links to communicate with the master communication device **102** or one or more service communication devices **106** for monitoring and control purpose while employs a laser beam deflection capability for actual laser beam based wireless communication in a multigigabit data rate to form a laser beam based wireless network, such as the second communication network **120**, in the defined indoor area **112**. Each of the plurality of optical routing devices **104** ensures an

ultra-flexible and ultra-reliable laser beam based wireless communication with significantly lower power consumption, which can be powered even by an inbuilt battery. The disclosed optical routing device, such as the optical routing device **10A**, creates a smart laser-based wireless network in the defined indoor area **112** that ensures full coverage with high signal-to-noise (SNR) in different physical spaces including all nooks and corners of the defined indoor area **112** with high-speed data connectivity to the plurality of end-user devices **108** without the need to deploy costly conventional intermediate routers. The alignment of the master communication device **102** with the one or more service communication device **106** may be done utilizing the plurality of optical routing devices **104** by use of a Z-fold arrangement of mirrors or alignment techniques, shown and discussed in detail, for example, in FIGS. **6A** and **6B**.

[0042] FIG. **2** is a block diagram illustrating various components of an exemplary master communication device of a communication system, in accordance with an exemplary embodiment of the disclosure. FIG. **2** is explained in conjunction with elements from FIG. **1**. With reference to FIG. **2**, there is shown a block diagram **200** of the master communication device **102**. The master communication device **102** may include a control section **202** and a front-end hybrid section **204**. The control section **202** may include a digital signal processor **206** (i.e., a DSP **206**) and a memory **208**. The control section **202** may be communicatively coupled to the front-end hybrid section **204**. The front-end hybrid section **204** may include a front-end radio frequency (RF) section **210**, a front-end optical section **212**, and a backhaul connection component **214**. The front-end RF section **210** may include front-end RF circuitry **216** and an RF antenna **218**. The front-end optical section **212** may include a laser handling system **220**, which includes a laser transmitter **222**, a laser detector **224**, a modulation circuit **226**, and a plurality of optical interfaces **228**.

[0043] The laser handling system **220** may further include a laser transmission (LT) circuitry **230** and a laser detector (LD) circuitry **232**. The LT circuitry **230** may include free-space laser transmitter electronics, such as a laser driver **234**, the modulation circuit **226**, and in some exemplary implementations, additionally one or more beam-shaping optics (not shown). The LD circuitry **232** may include free-space laser detector electronics, such as a transimpedance amplifier (TIA) for converting the current signal from the laser detector **224** to a voltage signal, a limiting amplifier for amplifying and cleaning the signal, and a Clock and Data Recovery (CDR) component for recovering clock and data signals from the incoming signal. Additionally, in some implementation, the LD circuitry **232** may further include other components, such as a pre-amplifier, a post-amplifier, and an equalizer in conjunction with the TIA, limiting amplifier, and the CDR to increase overall performance of the communication system **100** depending on the use case. There is further shown an optical wireless power transmission (OWPT) component **236** in the laser handling system **220**, for the purpose of remote delivery of power to the plurality of optical routing devices **104** and the one or more service communication devices **106**.

[0044] The DSP **206** include suitable logic, circuitry, and/or interfaces configured to control the laser handling system **220**. In an implementation, the DSP **206** may be further configured to control the front-end RF circuitry **216**. The master communication device **102** may be a programmable device, where the DSP **206** may execute instructions stored in the memory **208**. Example of the implementation of the DSP **206** may include, but are not limited to an embedded processor, a field-programmable gate array (FPGA), a microcontroller, a specialized DSP, an Application-Specific Integrated Circuit (ASIC) processor, and/or other processors.

[0045] The memory **208** may include suitable logic, circuitry, and/or interfaces that may be configured to store instructions executable by the DSP **206**. Examples of implementation of the memory **208** may include, but not limited to, a random access memory (RAM), a dynamic random access memory (DRAM), a static random access memory (SRAM), a processor cache, a thyristor random access memory (T-RAM), a zero-capacitor random access memory (Z-RAM), a read only memory (ROM), a hard disk drive (HDD), a secure digital (SD) card, a flash drive, cache memory, and/or other non-volatile memory. It is to be understood by a person having ordinary skill in the art

that the control section **202** may further include one or more other components, such as an analog to digital converter (ADC), a digital to analog (DAC) converter, a cellular modem, and the like, known in the art, which are omitted for brevity.

[0046] The front-end RF section **210** may include the front-end RF circuitry **216** and the RF antenna **218**. The front-end RF circuitry **216** may include RF receiver circuitry and RF transmitter circuitry for Wi-Fi® communication, for example, in 2.4 GHz, 5 GHz, 6 GHz, or 7 GHz. In an example, the receiver circuitry may include a cascading receiver chain comprising various components (e.g., the RF antenna **218**, a set of low noise amplifiers (LNA), a set of receiver front end phase shifters, and a set of power combiners) for the signal reception (not shown for brevity). Thus, the master communication device **102** may be a modified wireless access point (WAP) where at least one RF antenna is retained to communicate control signals to the plurality of optical routing devices **104** and the one or more service communication devices **106** to maintain service continuity in the laser beam based wireless communication network and where the front-end optical section **212** is connected at the other end to facilitate communication with different network nodes, such as the plurality of optical routing devices **104**, via laser beams. The RF antenna **218** may be an RF antenna of the modified WAP.

[0047] The laser handling system **220** may include the laser transmitter **222**, the laser detector **224**, the modulation circuit **226**, and the plurality of optical interfaces **228**. The laser transmitter **222** may be configured to direct a first laser beam carrying the first data signal over the air in a downstream path to a service communication device, such as the service communication device **106A**, directly or via the plurality of optical routing devices **104** based on the plurality of defined connectivity criteria. The laser transmitter **222** may be configured to focus the first laser beam into a narrow, collimated beam. The laser detector **224** may be configured to detect a second laser beam carrying one or more second data signals from the plurality of end-user devices **108** via the plurality of optical routing devices **104**. The laser detector **224** detects the photons of light that are emitted by the second laser beam and converts them into an electrical signal that can be analyzed and data can be extracted from such electrical signal. The modulation circuit **226** may be configured to modulate the intensity and/or any one of: a phase, frequency, or a polarization of the first laser beam (generated by the laser transmitter **222**) in accordance with the data to be transmitted. In an implementation, the laser transmitter **222** may include a laser diode that may be used to generate the first laser beam in which the first data signal may be modulated by the modulation circuit **226**.

[0048] Beneficially, the plurality of optical interfaces **228** are special purpose physical interfaces to handle a plurality of laser beams concurrently segregating download and upload functions via a plurality of distinct laser beams. There may be three scenarios for implementation of the plurality of optical interfaces **228**. In a first implementation, there may be one bi-directional optical interface **228A** to direct one laser beam for both downstream (Tx) and upstream communication (Rx) by wavelength division multiplexing (WDM). As an example, 1490 nm wavelength can be used for transmitting and 1550 nm for receiving. The task of beam combining and separating can be done with a dichroic mirror. In the second implementation, there may be two optical interfaces with one optical interface **228A** to direct one laser beam for downstream (Tx) and another optical interface **228B** for upstream communication (Rx). In a third implementation, there may be three optical interfaces **228A** and **228B**, where one optical interface **228A** is used to direct a bi-directional laser beam for laser-based wireless communication in a non-visible, near infrared light spectrum to form a wireless optical backhaul, and another optical interface **228B** is used to direct a beacon laser beam in the visible spectrum for the purpose of optical alignment, and the other optical interface **228C** is used for power transmission to downstream optical nodes, such as one or more optical routing devices of the plurality of optical routing devices **104** and/or the one or more service communication devices **106**. The DSP **206** may be further configured to control emission of a third laser beam (e.g., may also be referred to as a beacon laser beam or a power laser) via the optical

interface **228C** over-the-air for an optical wireless power transmission to remotely charge the rechargeable battery **314** (FIGS. 3A and 3C) of each of the plurality of optical routing devices **104** (e.g., passive optical deflectors or passive free-space optical routers for free-space communication (Not optical fiber routers)). For example, when the third laser beam strikes one of the plurality of optical routing devices **104**, the third laser beam may be deflected further to other optical routing devices causing each rechargeable battery of each of the plurality of optical routing devices **104** to be charged wirelessly via an optical medium. In an implementation, the optical interface **228C** may be placed in the middle of the other two optical interfaces **228A** and **228B**. In another implementation, the optical interface **228C** may be positioned nearest the mounting surface (e.g., the ceiling surface) in the order optical interface **228C** followed by optical interfaces **228A** and **228B** in a vertical alignment or a horizontal alignment or at other positions without limiting the scope of the disclosure.

[0049] The backhaul connection component **214** may include ports to connect to wired mediums, such as a coaxial cable, a fiber optic cable and the like, to receive input, for backhaul connectivity, for example, to Internet (i.e., the first communication network **118**), and provide output to the laser handling system **220**. The backhaul connection component **214** may be configured to convert the first data signal received as the optical signal or the RF signal to an electrical signal and pass data in the form of the electrical signal to the laser handling system **220** where it is modulated to a laser beam, such as the first laser beam, for over-the-air laser beam based wireless communication.

[0050] The LT circuitry **230** may include the laser driver **234**, the modulation circuit **226**, or one or more beam shaping optical components. The laser driver **234** may be configured to provide the electrical current needed to operate the laser transmitter **222**, such as a directly modulated laser diode (DML) or an externally modulated laser diode (EML). The laser driver **234** may be used to operate the laser transmitter **222** to direct a laser beam over-the-air within its specified current range, providing stable and reliable operation. The laser driver **234** drives the laser diode by converting a low-voltage, low-current signal into a high-voltage, high-current signal that can drive the laser diode. The one or more beam shaping optical components may be used to shape the transmitted laser beam to match the requirements of the communication system **100**, for example, by collimating the laser beam, expanding, focusing the laser beam, or adjusting the laser beam's divergence angle so that the laser beam is properly coupled into an optical communication channel and propagates with optimum spreading. The beam shaping optics may use a Keplerian design or preferably a Galilean design for compactness.

[0051] The LD circuitry **232** may be configured to detect and process the signals received from the laser detector **224** to achieve free-space optical communication. In an implementation, the LD circuitry **232** may employ laser detector electronics known in the art. The sub-components of the LD circuitry **232** are not shown for the sake of brevity. For example, the LD circuitry **232** may include the TIA, the limiting amplifier, and the CDR component for detecting and processing the signals received from the laser detector **224**. In an implementation, for example, the TIA may be used convert a current signal generated by the laser detector **224** into a voltage signal. The TIA may amplify the current signal by converting it into a voltage signal. The TIA may further provide a low-impedance output that can be easily connected to other components. The TIA may be a part of front-end amplifier that may be employed to boost the signal to a level that can be detected by the limiting amplifier. The limiting amplifier may then be used to amplify the voltage signal produced by the TIA and to remove any noise or distortion in the signal. The limiting amplifier may have a threshold that sets the upper and lower limits of the input signal. When the input signal is below the lower threshold, the limiting amplifier output is zero, and when the input signal is above the upper threshold, the limiting amplifier output is at its maximum level. The limiting amplifier may be employed to ensure that the signal is clean and that it is not distorted by any noise. The CDR component may be used to recover the clock signal and the data signal from the incoming signal. The clock signal may be used to synchronize the laser detector **224** with the laser transmitter from

where a laser beam with the first data signal is received receiver. In other words, to synchronize the receiver with the transmitter, while the data signal carries the actual information. The CDR component may be configured to utilize a phase-locked loop (PLL) to extract the clock signal from the incoming signal and to generate a local clock signal that is synchronized with the transmitter (e.g., the laser transmitter **420** of the service communication device **106A**). The data signal may then be extracted by sampling the input signal at the correct time relative to the recovered clock signal. The CDR component may be employed to ensure that the receiver (e.g., the laser detector **224**) is synchronized with the transmitter (e.g., (e.g., the laser transmitter **420** of the service communication device **106A**) and that the data is correctly decoded.

[0052] Additionally, in some implementation, the LD circuitry **232** may further include other components, such as a pre-amplifier, a post-amplifier, and equalizer in conjunction with the TIA, limiting amplifier, and the CDR to increase overall performance of the communication system **100**. For instance, in a case where the received signal is weak, and a stronger signal is needed to drive the TIA, the pre-amplifier may be used to amplify weak signals received by the laser detector **224** before it is passed to the TIA. In a case where the gain of the signal needs to be adjusted to match the requirements of the downstream components, the post-amplifier may be used to amplify the signal after it has passed through the limiting amplifier. Similarly, the equalizer may be used to compensate for any distortion, if any introduced in the communication system **100**.

[0053] The OWPT component **236** may be configured to direct a third laser beam (e.g., may also be referred to as a beacon laser beam or power laser) via the optical interface **228C** over-the-air for an optical wireless power transmission to remotely charge the rechargeable battery **314** (FIGS. **3A** and **3C**) of the optical routing device **104A** and further other the optical routing devices of the plurality of optical routing devices **104**. For example, when the third laser beam (may also be referred to as a power laser or a beacon laser beam) strikes one of the plurality of optical routing devices **104**, the third laser beam may be deflected further to other optical routing devices causing each rechargeable battery of each of the plurality of optical routing devices **104** to be charged wirelessly. Alternatively, in some implementations, a third laser beam via the optical interface **228C** may only be used for optical wireless power transmission over-the-air to remotely charge the rechargeable battery **314** (FIGS. **3A** and **3C**) of the optical routing device **104** and may not be used for alignment. A separate beacon laser beam in visible light may be used for alignment purpose via another optical interface, such as the optical interface **228B** in an example. In an implementation, the OWPT component **236** may be further configured to remotely charge the rechargeable battery **440** (FIG. **4**) of the service communication device **106A** and/or other service communication devices. In an implementation, the OWPT component **236** may be operably connected to the LT circuitry **230** and may have its own laser diode distinct from the laser transmitter **222** to function concurrently when in operation. In another implementation, the OWPT component **236** may be a transceiver (e.g., metal halide perovskite photovoltaic cell) that may be employed for both absorbing light from and emitting light (i.e., the third laser beam) to other OWPT systems equipped with similar transceivers, and an additional laser diode may not be required. The OWPT component **236** may be further configured to transmit the third laser beam (e.g., the power laser or the beacon laser beam) in a wavelength to match a spectral response of the laser energy harvesting cell **316A** of the optical routing device **104A**. The third laser beam (e.g., the power laser or the beacon laser beam) may be emitted in a defined power density to generate enough electricity (e.g., in milliwatts or even watts) to charge the rechargeable battery of other optical nodes, such as each of the plurality of optical routing devices **104** or the service communication device **106A**.

[0054] FIG. **3A** is a block diagram illustrating various components of an exemplary optical routing device of a communication system, in accordance with an exemplary embodiment of the disclosure. FIG. **3A** is explained in conjunction with elements from FIG. **1** and FIG. **2**. With reference to FIG. **3A**, there is shown a block diagram **300A** of the optical routing device **104A**. The optical routing device **104A** may be similar to that of other optical routing devices, such as the

plurality of optical routing devices **104**. Each of the plurality of optical routing devices **104** may be a light weight and a low power optical node with a small form factor. In an example, the dimensions of each of the plurality of optical routing devices **104** may range from 2 cm to 30 cm (width or length). In certain scenarios, the length may be extended using a length extender to maintain an alignment to send or receive a laser beam.

[0055] The optical routing device **104A** may include a mounting component **302** and an optical routing component **304** attached to the mounting component **302**. The mounting component **302** may be made of a variety of materials, including metal, mirror, or plastic, and may be sturdy and precise in order to accurately position the optical routing component **304**. The mounting component **302** may hold the optical routing component **304** and allows it to be positioned and oriented in one or more angles to deflect one or more laser beams in one or more specific directions and angles. In some cases, the optical routing component **304** may further include a free-space optical beam splitter (shown in FIG. 3C in an example) or a free-space optical beam combiner (shown in FIG. 3C in an example) depending on the application scenario. The mounting component **302** may be mounted on a ceiling surface or a wall surface of the defined indoor area **112**. In an example, mounting component **302** may be an electromechanical turret (also referred to as “eyeball”), or other mounting structure that is movable in 360 degrees around an axis (e.g., Y axis, and at least 30 degrees up and down with respect to an imaginary reference surface, running parallel to a ceiling surface in the defined indoor area **112**). In another example, a small stepper motor driven mechanism may be used in the mounting component **302** to pan and tilt the optical routing device **104A**. In another example, a small stepper motor driven mechanism may be provided in the mounting component **302** as a part of the movement controller **308** to pan and tilt the optical routing device **104A**.

[0056] The mounting component **302** may comprise a rechargeable battery **314** to power operations of the optical routing device **104A**. Beneficially, the optical routing device **104A** is developed to operate without the use of any external power supply from a power source (e.g., an alternating current power source), which in turn facilitates easy deployment of the optical routing device **104A**. Moreover, the way the rechargeable battery **314** is intelligently charged using the same laser beam that is employed to carry a data signal in the laser beam based wireless communication network (i.e., the second communication network **120**). This ensures flexibility in deployment as well as increases the reliability in operations of the optical routing device **104A**. The optical routing device **104A** may further include a laser energy harvesting cell **316A** and a charging handler **316B**.

[0057] The mounting component **302** may further comprise the processor **306** configured to communicate over-the-air with the master communication device **102** or one or more service communication devices **106** via one or more radio frequency (RF) supervisory links. In an example, the processor **306** may be implemented as low-power control circuit, which may be powered ON only when needed to save power. Other examples of the processor **306**, may include but are not limited to a control circuitry, a microcontroller, a microprocessor, a state machine, a hardware processor, and or other control circuits. In a case where the optical routing device **104A** is a primary optical deflector node, the processor **306** may be configured to establish an RF supervisory link with the master communication device **102**. In a case where the optical routing device **104A** is a secondary optical deflector node, the processor **306** may be configured to establish one or more RF supervisory links with one or more other optical routing devices of the plurality of optical routing devices **104** and/or one or more service communication devices **106**. The primary optical deflector node refers to any optical routing device that in a line-of-sight of the master communication device **102** and/or a nearest optical node that is able to send and receive laser beam from the master communication device **102**. Each secondary optical deflector node refers to an optical routing device that is in a non-line-of-sight (NLOS) to the master communication device **102** and/or not directly able to send and receive laser beam from the master communication device **102**. The processor **306** may be configured to communicate with the master

communication device **102** or one or more service communication devices **106** by use of the RF control signal handler **310**. For example, in an implementation, the RF control signal handler **310** may be configured to communicate control signals under the control of the processor **306** over existing Wi-Fi® signals in 2.4 GHz or 5-7 GHz frequency with the master communication device **102** to receive instructions in real time or near real time to change an angle or a direction of deflection of a laser beam in order to switch and align the laser beam to a different destination, such as a new optical routing device from among the plurality of optical routing devices **104**. In another implementation, the plurality of optical routing devices **104** may be controlled and networked via a built-in BLUETOOTH™ link. In other words, RF supervisory links may be Bluetooth links or Wi-Fi® links.

[0058] The processor **306** may be further configured to receive an instruction via the one or more RF supervisory links to control a movement of the mounting component **302** along with the optical routing component **304** such that an angle or a direction of deflection or routing of one or more laser beams from the optical routing component **304** is changed. In an example, the processor **306** may be configured to direct the movement controller **308** to move the mounting component **302** along with the optical routing component **304**. The movement controller **308** enables the optical routing component **304** to be positioned and oriented in one or more angles to reflect a laser beam in one or more specific directions and angles. In an example, the movement controller **308** may include a servo motor, a stepper motor, or an actuator.

[0059] The optical routing component **304** may comprise one or more laser beam handling regions **318** configured to handle a plurality of laser beams concurrently in which a first laser beam is deflected via a first region **318A** of the one or more laser beam handling regions **318** for downstream data communication in a downstream path and a second laser beam is deflected via a second region **318B** of the one or more laser beam handling regions **318** for upstream data communication in an upstream path. In an example, the one or more laser beam handling regions **318** may be two distinct deflection interfaces or laser beam handling interfaces provided on a deflecting surface **320** configured to deflect two laser beams concurrently, where one laser beam may be directed towards the downstream path and the other laser beam may be directed towards the upstream path. In accordance with an embodiment, the downstream path corresponds to communication from the master communication device towards the one or more service communication devices via at least the optical routing device **104A**. The upstream path corresponds to communication from one or more end-user devices to the one or more service communication devices and further to the master communication device via at least the optical routing device **104A**. Each of the one or more laser beam handling regions **318** may comprise a mirror or a mirror-like surface to deflect a corresponding laser beam. Each of the one or more laser beam handling regions **318** may include a mirror or a mirror-like surface to deflect one laser beam. For example, the mirror-like surface may be made from a variety of materials and coatings, such as aluminum, or silver, and polished and coated in order to achieve the highest possible reflectivity. In another example, each of the one or more laser beam handling regions **318** may include a free-space beam splitter or combiner, or a free-space wavelength splitter or combiner. In yet another implementation, one of the one or more laser beam handling regions **318** may be used for deflecting or routing a single laser beam for Tx and Rx (i.e., for bi-directional communication) but at two different wavelengths using wavelength division multiplexing (WDM). The wavelength splitting and combining may be performed passively using a dichroic mirror.

[0060] In accordance with an embodiment, the first laser beam may carry a first data signal from the master communication device **102**. The second laser beam may carry one or more second data signals from the one or more service communication devices **106**. The one or more second data signals correspond to user data associated with a plurality of end-user devices **108**.

[0061] In accordance with an embodiment, the optical routing device **104A** may use multiple optical wavelengths in deflected beams to send and receive data. For example, similar to a gigabit

passive optical network (GPON), the processor **306** may be further configured to use separate wavelengths for the first laser beam at the 1550 nm and the second laser beam at 1300 nm bands for transmit and receive respectively. Additional wavelength channels (i.e., communication data lanes) in each band may offer a path to increasing the data rate. For example, coarse wavelength division multiplexing (CWDM) may allow up to 18 channels and dense wavelength division multiplexing (DWDM) may allow up to 200 channels.

[0062] In accordance with an embodiment, the processor **306** may be further configured to activate an Optical Wireless Charging-Cum-Networking (OWCCN) mode in which at least one of the one or more laser beam handling regions **318** is configured to deflect a laser beam to form a laser beam-based wireless communication network and concomitantly cause the same laser beam to charge the rechargeable battery **314** via the at least one of the one or more laser beam handling regions **318**. The OWCCN mode is a new mode which when activated not only uses the same laser beam (e.g., the first or the second laser beam) to communicate data in the Laser Beam Mesh Network (LBMN) or the Laser Beam Cascaded Network (LBCN) in the defined indoor area **112** but also charge the rechargeable battery **314** using the same laser beam. An example of the Laser Beam Cascaded Network (LBCN) is shown and described, for example, in FIG. **6A**. An example of the Laser Beam Mesh Network (LBMN) is shown and described, for example, in FIG. **6B**.

[0063] In an exemplary scenario, the master communication device **102** may be further configured to direct the first laser beam carrying the first data signal in the downstream path to a service communication device, such as the service communication device **106A**, via the plurality of optical routing devices **104**. The service communication device **106A** may be further configured to obtain one or more second data signals from the plurality of end-user devices **108** and re-transmit the obtained one or more signals over the second laser beam in the upstream path to the master communication device **102** via the plurality of optical routing devices **104**. In such exemplary scenario, depending on the data traffic in the upstream path or the downstream path, one of the one or more laser beam handling regions **318** may be selected to not only deflect the laser beam to form the laser beam-based wireless communication network but also cause the same laser beam to charge the rechargeable battery **314** via the selected one of the one or more laser beam handling regions **318**. For example, the second region **318B** of the one or more laser beam handling regions **318** may be selected if data traffic in the downstream path is less than the upstream path if the second region **318B** is used for the downstream data communication in the downstream path.

[0064] In another example, the selection of the laser beam handling region from the one or more laser beam handling regions **318** for charging may be pre-defined and not dynamic. In such a case, an electrical connection with such laser beam handling region is pre-defined and fixed with the charging handler **316B**. When the laser beam strikes such laser beam handling region (e.g., a deflection region), such as the second region **318B**, the laser energy harvesting cell **316A** (e.g., a photovoltaic cell) associated with the charging handler **316B** may absorb the light energy and converts it into electrical energy, which is then stored in the rechargeable battery **314**.

[0065] In yet another example, the selection of the beam handling region from the one or more laser beam handling regions **318** for charging may be dynamic. It is understood that delivering high power to the laser energy harvesting cell **316A** can quickly increase their temperature up to 120 or 150 degree Celsius at which point they can only remain operational for a few seconds.

Advantageously, when the OWCCN mode is set and activated, the processor **306** may be further configured to intermittently enable and disable a connection of the selected beam handling region from the one or more laser beam handling regions **318** to the rechargeable battery **314**. Specifically, the connection with the laser energy harvesting cell **316A** by the charging handler **316B** may be intermittently disabled to prevent a rise of temperature at the laser energy harvesting cell **316A** beyond a defined threshold limit. This enables to avoid the use of any bulky heat sink for heat dissipation and maintain the small form factor of the optical routing device **104A**. In another example, alternatively, if the Status-of-charge (SoC) of the rechargeable battery **314** is below a set

SoC, then it may be required to continually charge the rechargeable battery **314**. In such a case, the processor **306** may be further configured to switch the connection to the rechargeable battery **314** from the one or more laser beam handling regions **318**. In other words, the connection of the first region **318A** to the rechargeable battery **314** via a first laser energy harvesting cell, such as the laser energy harvesting cell **316A**, may be disabled by the charging handler **316B**, whereas the connection of the second region **318B** to the rechargeable battery **314** via a second laser energy harvesting cell may be enabled by the charging handler **316B** and vice-versa periodically.

[0066] In an implementation, the mounting component **302** may include a memory **312**. The memory may be configured to store instructions related to activation or deactivation of the OWCCN mode **312A**. The memory **312** may further store sensor information **312B** acquired from the one or more sensors **322** and optical path options **312C**. The optical path options **312C** may be different fall-back laser-link options available to a given optical routing device to establish a laser beam connectivity. The fall-back laser-link options for each optical routing device may include location coordinates of each optical routing device and corresponding two or more neighbouring optical routing devices and laser beam alignment information, such as an angle of deflection of a laser beam, a direction of deflection of the laser beam, a beam width of the laser beam, and a laser intensity to be set for the laser beam, between two optical routing devices. The fall-back laser-link options for each of the plurality of optical routing devices **104** may be used to generate the plurality of different laser-communication route options. There may be different optical path options **312C** for upstream path and the downstream path. In an example, the optical path options **312C** may be historical data of previous connections done by each of the plurality of optical routing devices **104** after installation to their corresponding locations in the defined indoor area **112**. In another example, the optical path options **312C** may be proactively determine options to establish laser beam-based connections by each of the plurality of optical routing devices **104**. Alternatively, and additionally, the optical path options **312C** may be acquired from the cloud server **116**.

[0067] FIG. **3B** is a diagram illustrating an exemplary optical routing device of a communication system, in accordance with an exemplary embodiment of the disclosure. FIG. **3B** is explained in conjunction with elements from FIGS. **1**, **2**, and **3A**. With reference to FIG. **3B**, there is shown the optical routing device **104A**.

[0068] In an implementation, the master communication device **102** (FIG. **1**) may be further configured to transmit the first laser beam **326A** in a wavelength to match a spectral response of the laser energy harvesting cell **316A** (e.g., photovoltaic cells) in the charging handler **316B** of the optical routing device **104A**. In an example, the master communication device **102** may be further configured to transmit the first laser beam **326A** in a defined power density to generate enough electricity (e.g., in milliwatts or even watts) to charge the rechargeable battery **314**. The movement of the mounting component **302** with respect to the optical routing component **304** may be achieved using a movable attachment element **332** that movably connects the mounting component **302** with the optical routing component **304**. The movable attachment element **332** may be a ball shaped movable element, an oval shaped movable element, a pipe-shaped movable element, or a combination thereof.

[0069] In accordance with an embodiment, the control of the movement of the mounting component **302** along with the optical routing component **304** may comprise controlling a pitch rotation of the mounting component **302** along an x-axis **330A** that is parallel to a mounting surface **328** of the optical routing device **104A**. The control of the movement may be in response to the instruction received from the master communication device **102** or the cloud server **116** via the one or more RF supervisory links. The mounting component **302** along with the optical routing component **304** may be rotated along the x-axis **330A**. The control of the movement of the mounting component **302** along with the optical routing component **304** may further comprise controlling a yaw rotation of the mounting component **302** along a y-axis **330B** orthogonal to the x-axis **330A**. The control of the movement of the mounting component **302** along with the optical

routing component **304** may further comprise causing an alignment of an optical path among the master communication device **102** one or more other optical routing devices, and the one or more service communication devices **106**, based on the controlling of the pitch rotation or the yaw rotation of the mounting component **302**.

[0070] In accordance with an embodiment, the control of the movement of the mounting component **302** along with the optical routing component **304** further comprises controlling a roll rotation of the mounting component **302** along a z-axis **330C**, where causing the alignment of the optical path among the master communication device **102**, one or more other optical routing devices, and the one or more service communication devices **106**, may be further based on the controlling of the roll rotation in addition to the pitch rotation or the yaw rotation of the mounting component **302**. Further, the control of the movement of the mounting component **302** along with the optical routing component **304** further comprises aligning a transmit polarization of the one or more laser beams deflected by the optical routing component **304** in accordance with a destination device when an optical polarization multiplexing is utilized in data communication over a laser beam-based wireless communication network. The optical polarization multiplexing is used to transmit multiple data streams over a single laser beam by encoding the data onto different polarization states of light. The transmit polarization of the one or more laser beams deflected by the optical routing component **304** may be aligned by adjusting the orientation of the optical routing component **304** so that it deflects light with a specific polarization state in the transmission path, for example, the upstream path or the downstream path. The polarization of laser emission refers to the direction of the electric field oscillation of a given laser beam. The destination device may be one of the master communication device **102**, one or more other optical routing devices, or the one or more service communication devices **106**.

[0071] In accordance with an embodiment, the mounting component **302** may further comprise one or more sensors **322** (such as an accelerometer, an electronic compass, a gyroscope, a magnetometer, a geospatial positioning sensor, and the like) to measure an orientation and a level of the optical routing component **304** with respect to a reference surface **334** to maintain an alignment of an optical path with one or more of the master communication device **102**, one or more other optical routing devices, and the one or more service communication devices **106**. The one or more sensors **322** correspond to the one or more sensors **122** of FIG. 1. The reference surface **334** may be an imaginary reference surface running parallel to a ceiling surface, such as the mounting surface **328**, in the defined indoor area **112**. The level corresponds to a distance from the reference surface **334**. The processor **306** may be further configured to communicate the measured orientation, the level, and location data of the optical routing device **104A** over the one or more RF supervisory links to the master communication device **102** or the cloud server **116**.

[0072] In accordance with an embodiment, the optical routing device **104A** may further comprise a temperature sensor **336**. The processor **306** may be further configured to determine a thermomechanical-induced change in an alignment of an optical path of the optical routing device **104A** with one or more of the master communication device **102**, one or more other optical routing devices, and the one or more service communication devices **106**. The processor **306** may be further configured to execute an automatic alignment of the optical path when a temperature monitored by the temperature sensor **336** is outside a pre-set range. The automatic alignment may use a look up table containing the correction to be made by a motor (e.g., a stepper motor) at different temperatures. Such a look up table may be pre-defined and generated during calibration at the time of manufacturing (e.g., in a factory). Alternatively, in another example, such look up table may be generated dynamically or updated remotely by the cloud server **116**.

[0073] In accordance with an embodiment, the processor **306** may be further configured to receive a Laser Beam Network Control (LBNC) instructions from the master communication device **102** or the cloud server **116**. The processor **306** may be further configured to dynamically adjust an orientation of the one or more laser beam handling regions **318** in the deflecting surface **320** of the

optical routing component **304**, based on the LBNC instructions received from the master communication device **102** or the cloud server **116**. The instruction to control the movement of the mounting component **302** may be received from the master communication device **102** or the cloud server **116** via the one or more RF supervisory links. The processor **306** may be further configured to switch and re-align the one or more deflected laser beams **326A1** and **326B1** in a real-time or a near real-time to a different destination device based on a change in the angle or the direction of deflection or pointing of the one or more laser beams from the optical routing component **304**. For example, the one or more deflected laser beams **326A1** and **326B1** may be deflected to a new optical routing device from among the plurality of optical routing devices **104** or a new service device of the one or more service communication devices **106**. An example of the change in the laser beam-based communication route or switching from one optical routing device to another is shown and described, for example, in FIG. **6A**.

[0074] FIG. **3C** is a block diagram illustrating various components of an exemplary optical routing device of a communication system, in accordance with another exemplary embodiment of the disclosure. FIG. **3C** is explained in conjunction with elements from FIGS. **1**, **2**, **3A** and **3B**. With reference to FIG. **3C**, there is shown a block diagram **300B** of the optical routing device **104N**. The optical routing device **104N** may be similar to that of the optical routing device **104A** except that the optical routing device **104N** may further include one or more lens **338A** and **338B**, one or more beam-splitters **340A** and **340B**, and one or more polarizers **342A** and **342B**.

[0075] In this embodiment, the optical routing device **104N** may further comprise one or more lens **338A** and **338B** mounted on one or more laser beam handling regions **318** to cause one or more of: a) focus a laser beam deflected to a destination device, b) collimate the laser beam deflected to the destination device, or c) change a shape of the laser beam deflected to the destination device. In some implementations, each of the one or more lens **338A** and **338B** may be used to focus or collimate the laser beam, or to change the size or shape of the beam. Each of the one or more lens **338A** and **338B** may be made of a variety of materials, including glass, plastic, and quartz. In an implementation, a filter may be used to block or absorb certain wavelengths of light, which can be used to fine-tune the characteristics of a laser beam.

[0076] In accordance with an embodiment, the optical routing component **304** may further comprise one or more beam-splitters **340A** and **340B**. Each beam-splitter may be configured to split a laser beam of the plurality of laser beams into two or more laser beams or combine multiple laser beams into a single beam.

[0077] In an implementation, the one or more beam-splitters **340A** and **340B** may be fixedly mounted one or more laser beam handling regions **318** of the optical routing component **304**. In another implementation, the one or more beam-splitters **340A** and **340B** may be movably mounted in the mounting component **302** independent of the one or more laser beam handling regions **318**. The processor **306** may be configured to extend or retract the one or more beam-splitters **340A** and **340B** from the mounting component **302** based on the LBNC instructions received from the master communication device **102** or the cloud server **116**.

[0078] In an exemplary implementation, at least one of the one or more beam-splitters **340A** and **340B** may be used in a case where the Laser Beam Mesh Network (LBMN) is generated, and one laser beam may be required to be formed into two or more separate laser beams. The one or more beam-splitters **340A** and **340B** may be free-space optical beam-splitters may or may not be used in laser beam cascaded network (LBCN) as cascaded deflection of one laser beam among multiple optical nodes, such as the plurality of optical routing devices **104**, may be performed for downstream communication. Further, cascaded deflection of another laser beam for upstream communication may be sufficient to create the laser beam cascaded network (LBCN) using the one or more laser beam handling regions **318**. In an implementation, at least one beam-splitter may be mounted at the location of one of the one or more laser beam handling regions **318**, such as the first region **318A**. Alternatively, at least one beam-splitter may be mounted at the optical routing

component **304** at a specific position to maintain an alignment of an optical route for downstream path or an upstream path or both to perform free-space optical communication. Examples of the one or more beam-splitters **340A** and **340B** may include but are not limited to a cube beam-splitter (having two prisms joined together as a cube), a plate beam-splitter, a Dichroic mirror, a half-silvered mirror, a polarizing beam-splitter, or a micro-mirror cut in a specific shape such that when an laser beam strikes on the micro-mirror, the laser light splits in a defined ratio into two separate laser beams, or an arrangement of one or more specific shaped mirrors or prisms. In yet another example, one of the one or more laser beam handling regions **318**, such as the first region **318A**, may be filled with an optically active liquid (e.g., a chiral liquid) that has the ability to split a light beam, such as a laser beam into two separate laser beams. In yet another example, one of the one or more laser beam handling regions **318**, such as the first region **318A** may be made of crystal-like substance, such as calcite or quartz that may be cut in a specific shape to split a light beam, such as a laser beam into two or more separate laser beams. Moreover, the one or more beam-splitters **340A** and **340B** may be used in reverse to combine two different beams into a single one or a dedicated free-space optical beam combiner may be used. In an implementation, one of the one or more laser beam handling regions **318**, such as the second region **310B**, to combine two different beams into a single beam for upstream communication towards the master communication device **102**. In an example, the third region **318C** may be used to receive a third laser beam (e.g., beacon laser beam in a visible light spectrum, for example at around 800 nm) for an optical beam alignment and for the optical wireless charging of the rechargeable battery **314**. In an implementation, the one or more beam-splitters **340A** and **340B** may be one of: a free-space optical wavelength splitter or combiner (i.e., free-space optical wavelength combiner) configured to either split an incoming laser beam of a first wavelength into different wavelengths or combine a plurality of incoming laser beams of different wavelengths into a single laser beam of the first wavelength. [0079] In accordance with an embodiment, the optical routing component **304** may further comprise one or more polarizers **342A** and **342B**. The processor **306** may be further configured to control a polarization of the plurality of laser beams, such as the first laser beam **326A** and the second laser beam **326B**, via the one or more polarizers **342A** and **342B**. In an implementation, the one or more polarizers **342A** and **342B** may be used to control the polarization of the laser beam, which can be used to filter out unwanted reflections or to modify the properties of the laser beam, for example, to accommodate more data, using multiple polarizations. The one or more polarizers **342A** and **342B** may be a free-space optical beam polarizers. Polarized laser beams are light waves in which the vibrations occur in a single plane and the process of transforming unpolarized light into polarized light is known as polarization. Beneficially, each of the plurality of optical routing devices **104** may have two distinct laser beam handling regions, such as the first region **318A** and the second region **318B** (may also be referred to as deflection regions) to handle a pair of laser beams concurrently in which the first laser beam **326A** may be used for downstream data communication in the downstream path and the second laser beam **326B** may be used for upstream communication in the upstream path.

[0080] In an implementation, the optical routing device **104A** may be a passive optical node, where there is no amplification of signals in the deflected one or more laser beams. Alternatively, in another implementation, the optical routing device **104A** may be an active optical node, where optical amplification may be employed to boost the signal in order to overcome for power loss. For example, in the 1550 nm wavelength band, Erbium Doped Fiber Amplifiers (EDFA) may be used and for the 1300 nm wavelength band, Semiconductor Optical Amplifiers (SOA) may be employed.

[0081] The optical routing device **104A** may further include the laser energy harvesting cell **316A** and the charging handler **316B**. Examples of the laser energy harvesting cell **316A** may include, but is not limited to a photovoltaic cell that may handle visible light spectrum (400-800 nm wavelength), a dye-sensitized solar cell (DSSC), a plasmonic DDSC, or another type of laser

energy harvesting cell that employs special nanoparticle-embedded energy harvesting cells like perovskite solar cells (PSCs), upconverting nanoparticles (UCNPs) or Infrared-to-visible photon up-conversion systems may be employed that can generate electricity not only from visible but also infrared or ultraviolet light spectrum. Typically, silicon-based photovoltaic cells are efficient at converting light in the visible and near-infrared spectrum, while other materials, such as nanoparticle-embedded energy harvesting cells, PSCs, UCNPs based photovoltaic cells may be more efficient at converting light in other wavelengths of the spectrum. The generated power may be stored in an energy storage device, such as the rechargeable battery **314**. The charging handler **316** may be the optical wireless charging handler that may be configured to manage the power generated by the laser energy harvesting cell **316A** and may have safety mechanism to cut OFF an electrical connection between the laser energy harvesting cell **316A** and the rechargeable battery **314** to prevent over-charging and ensure a safe charging process. In such a case, the laser energy harvesting cell **316A** may be positioned at the third region **318C** (i.e., the laser beam handling region) to harvest the third laser beam (i.e., the power laser or the beacon laser beam) from the master communication device **102**. When the third laser beam strikes the laser beam handling region, such as the third region **318C**, the laser energy harvesting cell **316A** may absorb the light energy and convert it into electrical energy, which is then stored in the rechargeable battery **314**.

[0082] In operation, the master communication device **102** may be configured to obtain a first data signal from the data source **114A** or the modem **114B** (FIG. 1). The data source **114A** or the modem **114B** may be used to provide an Internet connection from an Internet service provider or from a core network of a telecommunication network. The first data signal may comprise wireless network data.

[0083] In accordance with an embodiment, the master communication device **102** may be configured to modulate the first data signal into the first laser beam **326A** (FIG. 3B). The first data signal may be modulated into the first laser beam **326A** using one or more of an intensity modulation, a phase modulation, a frequency modulation, and a polarization modulation. In intensity modulation, the intensity or power of the first laser beam **326A** may be varied in accordance with the data to be transmitted. For example, if the data to be transmitted is a digital signal with two levels (0 and 1), the laser beam can be turned ON and OFF at a high frequency to represent a “1” and left “ON” continuously for a defined period to represent a “0”. The master communication device **102** may include the laser handling system **220** that includes the laser transmitter **222** (e.g., a laser diode) and the modulation circuit **226**. The laser transmitter **222** (e.g., the laser diode) may be used to generate the first laser beam **326A**, and the modulation circuit **226** may be configured to modulate the intensity and/or any one of: a phase, frequency, or a polarization of the first laser beam **326A** in accordance with the data to be transmitted. The phase of the first laser beam **326A** refers to a relative timing of the peaks and troughs of the electromagnetic waves that make up the first laser beam **326A**. The phase of the first laser beam **326A** may be controlled and manipulated through various known techniques, such as using phase masks or introducing certain changes in the path that the first laser beam **326A** travels through. In another example, wavelength division multiplexing (WDM) may be employed to modulate data in the first laser beam **326A**. For instance, the first laser beam **326A** may be generated at a specific wavelength, where data modulated in the first laser beam **326A** at the specific wavelength may be extracted by a detector (i.e., a laser detector **422** of FIG. 4) of the service communication device **106A** that is sensitive to the specific wavelength of the first laser beam **326A**. In another example, multiple streams of data may be modulated and transmitted concurrently using WDM at different wavelengths to increase the capacity of the first laser beam **326A** that may be transmitted in a downstream path in a laser-based wireless communication link towards the service communication device **106A**.

[0084] In yet another example, a frequency shifting operation may be used to modulate the first

signal into the first laser beam **326A**. In this case, a frequency of the first laser beam **326A** may be shifted slightly up or down to encode data. The shift in frequency may be achieved using a variety of methods, such as modulating the current that drives the laser transmitter **222** of the laser handling system **220** in the master communication device **102**, or using a device called an acousto-optic modulator. In another example, a wavelength hopping operation may be performed, where the wavelength of the first laser beam **326A** may be rapidly switched between different values to encode data, such as the first signal. The switching may be executed, for example, by use of a device called a diffraction grating or a microelectromechanical system (MEMS) mirror. In yet another example, the polarization of the first laser beam **326A** may be used to encode data. For example, the first laser beam **326A** may be linearly polarized in different directions to represent different data bits. In yet another example, the intensity of the first laser beam **326A** may be modulated to encode the first signal (i.e., data signal). For example, the first laser beam **326A** may be turned on and off at different frequencies to represent different data bits.

[0085] The master communication device **102** may be further configured to direct the first laser beam **326A** carrying the first data signal in a downstream path to a service communication device, such as the service communication device **106A**, directly or via the plurality of optical routing devices **104** based on a plurality of defined connectivity criterions. In the present disclosure, the master communication device **102** do not require any intermediate routers or satellite routers in order to reach to remote end-user devices, such as the plurality of end-user devices **108**. In other words, data do not need to traverse through multiple intermediate routers to reach to the remote client devices, such as the plurality of end-user devices **108**. The master communication device **102** that includes the laser handling system **220** may further include the laser transmitter **222** that emits the first laser beam **326A** carrying the first data signal towards, for example, the optical routing device **104A**. The first laser beam may be emitted in a non-visible light spectrum when in operation. The optical routing device **104A** may deflect (e.g., like a mirror) the first laser beam **326A** carrying the first data signal to another optical routing device **104B**, without any processing of the received first laser beam **326A** carrying the first data signal. In other words, an angle, and a direction of the first laser beam **326A** changes when the first laser beam **326A** strikes the optical routing device **104A**. The optical routing device **104B** further deflects the first laser beam **326A** further towards the service communication device **106A** if the service communication device **106A** is in a line-of-sight (LOS) to the optical routing device **104B**. In a scenario where there is a direct LOS between the master communication device **102** and the service communication device **106A**, and there is no signal obstruction or signal attenuation entity between the master communication device **102** and the service communication device **106A**, the master communication device **102** may direct the first laser beam **326A** carrying the first data signal in a downstream path to the service communication device **106A** directly.

[0086] In accordance with an embodiment, the plurality of defined connectivity criterions may comprises a first criterion of an absence of a line-of-sight between the master communication device **102** and the service communication device **106A**, a second criterion of a presence of a signal obstruction entity or a signal attenuating entity between the between the master communication device **102** and the service communication device **106A**, and a third criterion of a presence of a signal-to-noise ratio (SNR) less than a defined threshold number of decibels at the second location of the service communication device **106A** when test radio frequency (RF) signals are communicated from the master communication device **102**. In such plurality of defined connectivity criterions, the master communication device **102** directs the first laser beam **326A** carrying the first data signal to the service communication device **106A** via the plurality of optical routing devices **104**.

[0087] In accordance with an embodiment, when one or more of the first criterion, the second criterion, or the third criterion are met, the master communication device **102** may be further configured to determine the optical routing device **104A** among the plurality of optical routing

devices **104** to be selected to direct the first laser beam **326A** carrying the first data signal in the downstream path, and calibrate an angle of projection of the first laser beam **326A** from a reference point of the master communication device **102** to direct the first laser beam **326A** carrying the first data signal in the downstream path towards the determined optical routing device **104A**. The selection of the optical routing device **104A** may be made based on a direct LOS of the optical routing device **104A** from the master communication device **102**. Other factors that may be considered include an alignment with the downstream path to maintain the alignment of the first laser beam **326A**, and a beam width handling capability of an optical routing device. Although a wider beam may be used to transmit more data but may also be more susceptible to interference. Thus, a narrow pencil laser beam may be used in the form of the first laser beam **326A** to generate laser beam-based communication links between two network nodes (i.e., starting from the master communication device **102** to the optical routing device **104A**, followed by a laser beam-based communication link between two optical routing devices, and other successive optical routing devices until it reaches to the service communication device **106A** in the downstream path) except for communication with end-user devices, such as the plurality of end-user devices **108**.

[0088] The service communication device **106A** may be configured to detect the first laser beam **326A** carrying the first data signal in the downstream path. The service communication device **106A** may include a laser detector (the laser detector **422** of FIG. 4) to detect the first laser beam **326A** carrying the first data signal in the downstream path. In an example, the laser detector may be implemented as a photodiode or a phototransistor to detect the presence of the first laser beam **326A** by converting the optical energy into an electrical current. In some implementation, an optical filter may be used to filter out any unwanted light, allowing only the desired laser wavelength to pass through to the laser detector to improve the signal-to-noise ratio and increase the accuracy of the detection. In some implementation, a feedback may be generated by the service communication device **106A** by measuring a strength of the first laser beam **326A** allowing for the detection of the presence of first laser beam **326A** even if it is slightly misaligned with the laser detector. In such a case, the feedback may be used to automatically select another optical routing device from among the plurality of the optical routing devices **104** to maintain the laser beam based wireless connectivity and service continuity with the plurality of end-user devices **108**. The feedback may be communicated to the master communication device **102** which then communicates the feedback to the cloud server **116**. Alternatively, the feedback may be communicated to the cloud server **116** directly, for example, using radio frequency (RF) communication network (a fallback option) that may use existing RF-based Wi-Fi® network.

[0089] The service communication device **106A** may be further configured to demodulate the first data signal from the first laser beam **326A** and distribute one or more wireless signals to the plurality of end-user devices **108**. The demodulation may depend on the modulation technique used during the modulation of the first data signal from the first laser beam **326A**. The demodulation of the first data signal from the first laser beam **326A** may be carried out using one or more known demodulation methods in free-space optics communication. For example, using direct detection, a coherent detection, a Frequency-shift keying (FSK) demodulation, a Phase-shift keying (PSK) demodulation, an Amplitude shift keying (ASK) demodulation, and the like. In the direct detection, the first laser beam **326A** may be detected directly by the laser detector (e.g., a photodiode), where the output of the photodetector is then amplified and filtered to extract the first data signal (i.e., the data signal). For example, data may be encoded directly onto the intensity of the first laser beam **326A** during modulation, where intensity of the laser beam is modulated to represent the binary 1s and 0s of the data. To demodulate the data, the laser detector may be used to convert the optical signal into an electrical signal, which is then processed to extract the data. In the coherent detection, the received first laser beam **326A** may be mixed with a local oscillator (LO) beam of the same frequency, and the resulting beat frequency may be detected and demodulated to extract the first data signal (i.e., the data signal). In FSK demodulation, as the data may be encoded by shifting

the frequency of the laser beam between two values, typically referred to as “mark” and “space” frequencies, a frequency discriminator can be used to demodulate the FSK-modulated laser beam and extract the data. In PSK demodulation, as the data may be encoded by shifting the phase of the first laser beam **326A** between two or more values, a phase-locked loop (PLL) may be used to demodulate the PSK-modulated first laser beam **326A** and extract the data. In ASK, the data may be encoded onto the amplitude of the first laser beam **326A**, where amplitude of the first laser beam **326A** may be modulated to represent the binary 1s and 0s of the data. In such a case, to demodulate the data, the laser detector may be used to convert the optical signal into an electrical signal, which is then processed to extract the data.

[0090] The service communication device **106A** may be further configured to obtain one or more second data signals from the plurality of end-user devices **108** and re-transmit the obtained one or more signals over a second laser beam in an upstream path to the master communication device **102** directly or via the plurality of optical routing devices **104** based on the plurality of defined connectivity criteria. The one or more second data signals may be data signals received over RF waves from the plurality of end-user devices **108**. The service communication device **106A** may be further configured to extract user data (i.e., data signals) and re-modulate the extracted user data into the second laser beam, which is then transmitted to the master communication device **102**. The first laser beam **326A** and the second laser beam form a laser backhaul in the defined indoor area **112** controlled by the master communication device **102** in coordination with the plurality of optical routing devices **104** and one or more service communication devices **106**, such as the service communication device **106A**. Advantageously, instead of a single laser beam handling both downstream and upstream communication that require certain synchronizations, such as time and/or frequency synchronization to route data from source to destination and back from destination to the source, the communication system **100** bifurcates the downstream path and the upstream path physically, i.e., by having two different laser beams (the first laser beam **326A** and the second laser beam **326B**). Thus, if one wavelength of the first laser beam **326A** is employed for downstream path (e.g., for download) and another wavelength of the second laser beam is employed for routing back towards the master communication device **102**, then there is even no need to have time division duplexing (TDD).

[0091] In accordance with an embodiment, the first data signal from the first laser beam **326A** may be demodulated by the service communication device **106A** and re-modulated into a plurality of different radio frequency (RF) signals associated with different communication protocols having different communication ranges (e.g., BLUETOOTH™, Wi-Fi®™, Li-Fi, a citizens broadband radio service (CBRS) signal, and/or a wireless personal area network (WPAN) signal) to serve the plurality of end-user devices **108** in accordance with their wireless communication capabilities. The service communication device may be further configured to pack the content received from the plurality of end-user devices **108** and transmit upstream over the second laser beam to the master communication device **102** via the plurality of the optical routing devices **104**.

[0092] In one implementation, each of the master communication device **102**, the plurality of optical routing devices **104**, and the one or more service communication devices **106**, may have two distinct interfaces to handle two separate laser beam lights or routes. In accordance with an embodiment, each of the master communication device **102** and the service communication device **106A** may comprise two distinct physical communication interfaces to handle a pair of laser beams concurrently in which the first laser beam **326A** is used for downstream data communication in the downstream path and the second laser beam is used for upstream communication in the upstream path. Further, beneficially, each of the plurality of optical routing devices **104** may have two distinct deflection regions or laser beam handling regions to handle a pair of laser beams concurrently in which the first laser beam **326A** is used for downstream data communication in the downstream path and the second laser beam is used for upstream communication in the upstream path. In accordance with an embodiment, the master communication device **102** may be further

configured to exchange a plurality of data streams with the plurality of end-user devices **108** via a combination of the plurality of optical routing devices **104** and one or more service communication devices **106**.

[0093] In accordance with an embodiment, the master communication device **102** may be further configured to generate a Laser Beam Mesh Network (LBMN) or a Laser Beam Cascaded Network (LBCN) using the plurality of optical routing devices **104** disposed at the plurality of locations in the defined indoor area **112** to wirelessly connect to one or more service communication devices **106** in the defined indoor area **112**. An example of the Laser Beam Mesh Network (LBMN) is shown and described, for example, in FIG. **6B**. An example of the Laser Beam Cascaded Network (LBCN) is shown and described, for example, in FIG. **6A**.

[0094] In accordance with an embodiment, the master communication device **102** may be further configured to monitor performance data in terms of data rate, signal-to-noise ratio (SNR), a number of end-user devices connected to the service communication device and dynamically re-configure the generated Laser Beam Mesh Network (LBMN) or the Laser Beam Cascaded Network (LBCN). The data rate and the SNR of the one or more service communication devices **106** may be measured by the master communication device **102**. In case of any sudden performance drop, the master communication device **102** may be further configured to dynamically re-configure the generated LBMN or the LBCN to maintain service continuity with the plurality of end-user devices **108**, such as the UEs **108A** and **108B**.

[0095] In accordance with an embodiment, the master communication device **102** may be further configured to periodically communicate such performance data of the laser beam-based wireless communication network (i.e., the second communication network **120**) to the cloud server **116**. In an example, transmit laser power from laser transmitters as well as the optical receiver's received signal strength indicator (RSSI) for RF waves and laser power at the laser detectors at the one or more service communication devices **106** may be monitored. If transmit laser power has not degraded but the optical RSSI or laser power at the laser detectors indicates a drop in power, then an automatic optical alignment routine may be executed. The master communication device **102** may be further configured to receive the laser beam network control (LBNC) instructions from the cloud server **116**. Thereafter, the master communication device **102** may be further configured to cause at least one optical routing device (e.g., the optical routing device **104B**) of the plurality of optical routing devices **104** to dynamically adjust an orientation of a deflecting surface **320** of the at least one optical routing device, based on the received LBNC instructions from the cloud server **116**. Based on the received LBNC instructions from the cloud server **116**, the master communication device **102** may be further configured to dynamically change a laser beam-based communication route from the master communication device **102** to the service communication device **106A** by changing a path of laser communication from a first set of optical routing devices to a second set of optical routing devices of the plurality of optical routing devices **104** to reach to the service communication device **106A**. An example of the change in the laser beam-based communication route or switching from one optical routing device to another is shown and described, for example, in FIG. **6A**.

[0096] In accordance with an embodiment, the service communication device **106A** may be further configured to demodulate the first data signal from the first laser beam **326A** and re-modulate into a plurality of different radio frequency (RF) signals associated with different communication protocols having different communication ranges to serve the plurality of end-user devices **108** in accordance with their wireless communication capabilities. In some implementations, the service communication device **106A** may include not only an antenna for Wi-Fi® signals communication but also for communication in other personal area network (PAN) signals, such as BLUETOOTH™, a citizens' broadband radio service (CBRS) signal, and/or a wireless personal area network (WPAN) signals. In accordance with an embodiment, the service communication device **106A** may be further configured to pack the content received from the plurality of end-user

devices **108** and transmit upstream over the second laser beam (e.g., using WDM) to the master communication device **102**.

[0097] FIG. **4** is a block diagram illustrating various components of an exemplary service communication device of a communication system, in accordance with an exemplary embodiment of the disclosure. FIG. **4** is explained in conjunction with elements from FIGS. **1**, **2**, and **3**. With reference to FIG. **4**, there is shown a block diagram **400** of the service communication device **106A**. The service communication device **106A** may include a control section **402** and a front-end hybrid section **404**. The control section **402** may include a DSP **406**, a memory **408**, and the one or more sensor **122** (of FIG. **1**). The control section **402** may be communicatively coupled to the front-end hybrid section **404**. The front-end hybrid section **404** may include a front-end RF section **410** and a front-end optical section **412**. The front-end RF section **410** may include front-end RF circuitry **414** and a RF antenna **416**. The front-end optical section **412** may include a laser handling system **418**, which includes a laser transmitter **420**, a laser detector **422**, a modulation circuit **424**, a demodulation circuit **426**, and a plurality of optical interfaces **428**. In an implementation, the laser handling system **418** may further include a laser transmission (LT) circuitry **430** and a laser detector (LD) circuitry **432**. The LT circuitry **430** may include free-space laser transmitter electronics, such as a laser driver **434**, the modulation circuit **424**, and in some exemplary implementations, additionally one or more beam-shaping optics (not shown) similar to the master communication device **102**. The LD circuitry **232** may include free-space laser detector electronics, such as the TIA for converting the current signal from the laser detector **422** to a voltage signal, a limiting amplifier for amplifying and cleaning the signal, and a Clock and Data Recovery (CDR) component for recovering clock and data signals from the incoming signal similar to that of the master communication device **102**. The different components of the service communication device **106A** maybe similar to that of the master communication device **102** expect that the service communication device **106A** may not include a backhaul connection component, such as the backhaul connection component **214** of FIG. **2**, and further the demodulation circuit **426** may be a dual-function circuit that not only demodulates the data signal from the first laser beam but also performs mapping of the extracted data to a RF wave to be emitted by the RF antenna **416** to the plurality of end-user devices **108** and further extracts data received from the plurality of end-user devices **108** via one or more RF waves in upstream path and passes to the modulation circuit **424**. [0098] Beneficially, the plurality of optical interfaces **428**, especially the optical interfaces **428A** and **428B**, are special purpose physical interfaces to handle a plurality of laser beams concurrently segregating download and upload functions via a plurality of distinct laser beams. An optical interface, such as an optical interface **428A**, may be coupled to the laser detector **422** to receive a focused narrow beam over the air, for example, from one of the plurality of optical routing devices **104**. Another optical interface, such as an optical interface **428B**, may be coupled to the laser transmitter **420** to focus the second laser beam in a defined beam width, such as into a narrow, collimated beam, for example, to the one of the plurality of optical routing devices **104**. Further, another optical interface, such as an optical interface **428C**, may be coupled to the laser energy harvesting cell **438A** connected to the rechargeable battery **440** to charge the rechargeable battery **440** of the service communication device **106A**. The optical interface **428C** may also be used for optical beam alignment. Similar to that of the optical routing device **104A**, the service communication device **106A** may further include the charging handler **438B**. In other words, in some implementation, the service communication device **106A** may include the OWP section **436** that may include the laser energy harvesting cell **438A**, the charging handler **438B**, and the rechargeable battery **440** like the optical routing device **104A**. In some implementations, the laser energy harvesting cell **438A** may be a transceiver (e.g., metal halide perovskite transceiver) that may be employed for both absorbing light from and emitting light to other OWPT systems equipped with similar transceivers, and an additional laser diode may not be required. In an implementation, the DSP **406** may be further configured to dynamically adjust an orientation of the

plurality of optical interfaces **428** of the service communication device **106A**, based on the LBNC instructions received from the master communication device **102** or the cloud server **116**.

[0099] FIG. 5 is a block diagram illustrating different components of an exemplary cloud server, in accordance with an embodiment of the disclosure. FIG. 5 is explained in conjunction with elements from FIGS. 1, 2, 3A-3C, and 4. With reference to FIG. 5, there is shown a block diagram **500** of the cloud server **116**. The cloud server **116** may include a processor **502**, a network interface **504**, and a primary storage **506**. The primary storage **506** may further include a laser beam network controller **508** and a Laser Beam Network Control (LBNC) instructions generator **510**. There is further shown a machine learning model **512** and a routing database **514**. In an implementation, the cloud server **116** may be configured to acquire sensor data from the one or more sensors **122** through the master communication device **102** over the first communication network **118** (FIG. 1).

[0100] The routing database **514** may include one or more routing tables that stores a plurality of different laser-communication route options specific for each service communication device of the one or more service communication devices **106**. Similarly, it also stores fall-back laser-link options for each optical routing device of the plurality of optical routing devices **104**. The fall-back laser-link options for each optical routing device may include location coordinates of each optical routing device and corresponding two or more neighbouring optical routing devices and laser beam alignment information, such as an angle of deflection of a laser beam, a direction of deflection of the laser beam, a beam width of the laser beam, and a laser intensity to be set for the laser beam, between two optical routing devices. The fall-back laser-link options for each of the plurality of optical routing devices **104** may be used to generate the plurality of different laser-communication route options. There may be different look-up tables used for upstream path and the downstream path.

[0101] In an implementation, the processor **502** may determine the fall-back laser-link options for each of the plurality of optical routing devices **104** and the plurality of different laser-communication route options based on at least location coordinates of the master communication device **102**, the one or more service communication device **106**, and the plurality of optical routing devices **104**. The processor **502** may be further configured to use the machine learning model **512** to determine the fall-back laser-link options for each of the plurality of optical routing devices **104** and the plurality of different laser-communication route options based on at least location coordinates and environment mapping data of the defined indoor area **112**. The environment mapping data may be indicative of a 3D physical structure of the defined indoor area **112**, such as a 3D layout of an apartment or an enterprise.

[0102] In accordance with an embodiment, the processor **502** may be configured to cause the master communication device **102** to generate a Laser Beam Mesh Network (LBMN) or a Laser Beam Cascaded Network (LBCN) using the plurality of optical routing devices **104** disposed at the plurality of locations in the defined indoor area **112** to wirelessly connect to one or more service communication devices **106** in the defined indoor area **112**. Such generation of the Laser Beam Mesh Network (LBMN) or a Laser Beam Cascaded Network (LBCN) may be executed based on the sensor data obtained from the one or more sensors **122** at each optical node directly or via the master communication device **102** over the first communication network **118** (FIG. 1). An example of the LBMN and the LBCN is shown and described, for example, in FIGS. 6A and 6B. The processor **502** may be further configured to obtain performance data in terms of data rate, signal-to-noise ratio (SNR), a number of end-user devices connected to the one or more service communication devices **106**, from the master communication device **102**. The processor **502** may be further configured to cause the master communication device **102** to dynamically re-configure the generated Laser Beam Mesh Network (LBMN) or the Laser Beam Cascaded Network (LBCN) based on a change in the performance data that is beyond threshold. The changes, such as a drop in the performance data may be because of a default in alignment of laser beams between two optical routing devices or between one optical routing device and a service communication device, such as

the service communication device **106A**, or a new signal obstruction entity obstructs or interferes with the laser beam path.

[0103] In accordance with an embodiment, the processor **502** may be further configured to communicate LBNC instructions from the cloud server **116** to the master communication device **102**. Thereafter, the processor **502** causes the master communication device **102** to instruct at least one optical routing device (e.g., the optical routing device **104B**) of the plurality of optical routing devices **104** to dynamically adjust an orientation of a deflecting surface **320** of the at least one optical routing device, based on the communicated LBNC instructions. The LBNC instructions may be generated by the LBNC instructions generator **510** in coordination with the laser beam network controller **508**. The laser beam network controller **508** is configured to monitor, manage, and remotely control one or more network management functions of the laser beam based wireless communication network (i.e., the second communication network **120**) to enhance the network coverage area, bandwidth, reliability, and configurability in which a reconfigurable optical wireless backhaul is achieved. In accordance with an embodiment, the processor **502** may be further configured to cause the master communication device **102** to dynamically change a laser beam-based communication route from the master communication device **102** to the service communication device **106A** by changing a path of laser communication from a first set of optical routing devices to a second set of optical routing devices of the plurality of optical routing devices **104** to reach to the service communication device **106A**. Such a change in the laser beam-based communication route may be controlled based on a periodic communication of the LBNC instructions. In an implementation, the processor **502** may be configured to utilize the determined fall-back laser-link options for each of the plurality of optical routing devices **104** and the plurality of different laser-communication route options to cause the master communication device **102** to dynamically change the laser beam-based communication route.

[0104] FIG. **6A** is a diagram illustrating an exemplary implementation of a communication system with optical routing devices for ultra-flexible and ultra-reliable laser beam based wireless communication in an indoor area, in accordance with an exemplary embodiment of the disclosure. FIG. **6A** is explained in conjunction with elements from FIGS. **1-5**. With reference to FIG. **6A**, there is shown an exemplary scenario of a Laser Beam Cascaded Network (LBCN) **600A** generated using the plurality of optical routing devices **104**, such as the optical routing devices **104A**, **104B**, or **104C** in this case.

[0105] In a first scenario **602A**, the master communication device **102** is configured to obtain a first data signal from the modem **114B** via a wired medium, such as an optical fibre. The master communication device **102** may be a modified and intelligent wireless access point that uses existing Wi-Fi® signal in RF medium to control the optical routing devices **104A** and **104B** and employs laser beams to form the LBCN **600A** and communicate data in a cascaded structure via two distinct laser beams to reach to the service communication device **106A** (i.e., another modified wireless access point). In the LBCN **600A**, the plurality of optical routing devices **104**, such as the optical routing devices **104A** and **104B** are connected in series between the master communication device **102** and the service communication device **106A**, as shown, in an example. The mounting component **302** of some of the optical routing devices, such as the optical routing device **104A**, may be mounted on a ceiling surface **608**. The mounting component **302** may be movable in 360 degrees around an axis (e.g., Y axis, a vertical axis with respect to the ceiling surface **608**), and at least 30 degrees up and down with respect to the ceiling surface **608**. The optical routing device **104C** may be mounted on a wall surface **610A**.

[0106] In the first scenario **602A**, there is an absence of a line-of-sight between the master communication device **102** and the service communication device **106A**. Further, there is a presence of a signal obstruction entity **610** between the master communication device **102** and the service communication device **106A**. Furthermore, there is a presence of a signal-to-noise ratio (SNR) less than a defined threshold number of decibels is detected at the second location of the

service communication device **106A** when test radio frequency (RF) signals are communicated from the master communication device **102** via RF waves (e.g., one of the RF antenna **218**) in omni-directions. In this case, the master communication device **102** may be configured to determine the optical routing device **104A** among the plurality of optical routing devices **104** to be selected to direct the first laser beam **604** carrying the first data signal in the downstream path **612**. The master communication device **102** may be configured to calibrate an angle of projection of the first laser beam **604** from a reference point of the master communication device **102** to direct the first laser beam carrying the first data signal in the downstream path **612** towards the determined optical routing device **104A**. In this case, the reference point of the master communication device **102** may be an axis orthogonal to the optical interface **228A**. Thus, the master communication device **102** via the laser transmitter **222** may be further configured to direct the first laser beam **604** carrying the first data signal in the downstream path **612** for downstream communication to the service communication device **106A** via the optical routing devices **104A** and **104B**. The laser detector **422** of the service communication device **106A** may be configured to detect the first laser beam **604** carrying the first data signal in the downstream path **612**. The demodulation circuit **426** of the service communication device **106A** may be configured to demodulate the first data signal from the first laser beam **604**, and the RF antenna **416** may be configured to distribute the one or more wireless RF signals (e.g., in 5 GHz, 6 GHz, or 7 GHz frequency) to the UEs **108A** and **108B** after re-modulating the demodulated first data signal into the one or more wireless RF signals. Thereafter, the service communication device **106A** (e.g., via the RF antenna **416**) may be further configured to obtain one or more second data signals (i.e., RF signals) from the UEs **108A** and **108B** and re-transmit over a second laser beam **606** in an upstream path **614** to the master communication device **102** via the optical routing devices **104A** and **104B**. The DSP **406** or the demodulation circuit **426** (FIG. 4) may be configured to extract user data from the one or more second RF signals, and the modulation circuit **424** may then re-modulate the extracted user data into the second laser beam **606**. The second laser beam **606** may be directed by the laser transmitter **420** (FIG. 4) from the optical interface **428B** under the control of the DSP **406** towards the optical routing device **104B**, as shown, which is then deflected from the second region **310B** of the optical routing device **104B** to the second region **310B** of the optical routing device **104A** via the LBCN **600A** to reach the master communication device **102** for upstream communication.

Advantageously, a pair of laser beams such as the first laser beam **604** and the second laser beam **606** may be concurrently operated in which the first laser beam **604** is used for downstream data communication in the downstream path **612** and the second laser beam **606** is used for upstream communication in the upstream path **614**. Similarly, the optical routing component **304** may comprise the one or more laser beam handling regions **318** configured to handle a plurality of laser beams concurrently in which the first laser beam **604** may be deflected via the first region **318A** of the one or more laser beam handling regions **318** for downstream data communication in the downstream path **612** and a second laser beam **606** is deflected via the second region **318B** of the one or more laser beam handling regions **318** for upstream data communication in the upstream path **614**. In this case, the one or more laser beam handling regions **318** may be a plurality of deflection regions, as only deflection is performed passively without the need to perform any signal processing. Thus, a full coverage with high SNR (e.g., greater than 40 decibels) in different physical spaces is ensured including all nooks and corners of the defined indoor area **112** with high-speed data connectivity to end-user devices, such as the UEs **108A** and **108B** without the need to deploy costly and ineffective conventional intermediate routers with limited bandwidth that process signals, for example, for signal amplification. The concurrent handling of the pair of laser beams such as the first laser beam **604** and the second laser beam **606** for distinct but smooth downstream and upstream data communication improves data transfer rates (e.g., in double digit gigabit rate or even terabyte rate) between at least two remote wireless communication devices (such as two WAPs) in indoor scenario as compared to existing wireless systems (e.g., conventional

wireless local area networks).

[0107] Furthermore, the master communication device **102** may be configured to monitor performance data in terms of data rate, signal-to-noise ratio (SNR), a number of end-user devices connected to the service communication device **106A**. The master communication device **102** may be further configured to periodically communicate such performance data of the LBCN **600A** to the cloud server **116**. In an example, transmit laser power from laser transmitters (e.g., the laser transmitter **222** of FIG. **2** and the laser transmitter **420** of FIG. **4**) as well as the receiver optical received signal strength indicator (RSSI) and laser power at the laser detector **422** at the service communication device **106A** may be monitored. If transmit laser power has not degraded but the optical RSSI or laser power at the laser detector **422** indicates a drop in power, then an automatic optical alignment routine is executed.

[0108] In a second scenario **602B**, there may be some changes in the monitored performance data. The changes, such as a drop in the performance data may be because of a default in alignment of laser beams between the optical routing device **104B** and the service communication device **106A**, or when the laser power has not degraded but the RSSI indicates a drop in power, or a new signal obstruction entity **616** obstructs or interferes with the laser beam path. In such scenarios, a laser beam-based communication route may be dynamically changed from the master communication device **102** to the service communication device **106A** by changing a path of laser communication from a first set of optical routing devices **104A** and **104B** to a second set of optical routing devices **104A**, **104B**, and **104C** to reach to the service communication device **106A**. This is done to maintain service continuity to the UEs **108A** and **108B**.

[0109] FIG. **6B** is a diagram illustrating another exemplary implementation of a communication system with optical routing devices for ultra-flexible and ultra-reliable laser beam based wireless communication in an indoor area, in accordance with an exemplary embodiment of the disclosure. FIG. **6A** is explained in conjunction with elements from FIGS. **1-5**. With reference to FIG. **6A**, there is shown an exemplary scenario of a Laser Beam Mesh Network (LBMN) **600B** generated using the optical routing devices **104A**, **104B**, **104C**, **104D**, and **104E**. In this implementation, there are multiple service communication devices, such as the service communication devices **106A** and **106B**, each servicing its users, such as the UE **108A** and the UE **108B**, respectively.

[0110] Unlike a typical Wi-Fi® system, where there is only one broadcast device and hence a limited coverage area, there are certain other conventional Wi-Fi® systems that provide comparatively wider coverage by broadcasting from the main router to several satellite routers, which then broadcast to its client devices. The communication between the main router connected to the Internet and the satellite routers fed by the main router may be carried by the Backhaul. In order for the backhaul to not compete with the client devices for network bandwidth, one Wi-Fi® band is typically dedicated to the backhaul in such conventional systems. The reach and performance of these conventional systems and topology are fundamentally limited by the bandwidth of the dedicated band, and by the number of intermediate satellite routers that the data must traverse in order to reach remote client devices. While the backhaul connections can be done with physical cables which offer wider bandwidth than a Wi-Fi® band, cable installation and reconfiguration are extremely costly and slow. Further, conventional mesh Wi-Fi® system performance is heavily affected by the bandwidth of the backhaul and how the system manages the communication between the mesh router and its satellites, i.e., the satellite routers or intermediate routers.

[0111] In contrast to the conventional systems, the master communication device **102** may be further configured to cause the optical routing devices **104A**, **104B**, **104C**, **104D**, and **104E** to dynamically adjust the orientation of the deflecting surface **320** (which includes the one or more laser beam handling regions **318**) of the optical routing component **304** to form the LBMN **600B**.

[0112] In an implementation, the LBMN **600B** may be generated based on instructions received from the cloud server **116**. The master communication device **102** via the laser transmitter **222** may

be further configured to direct the first laser beam **604** carrying the first data signal in the downstream path **612** for downstream communication to the service communication devices **106A** and **106B** via the LBMN **600B**. A laser beam-based communication route in the downstream path **612** and the upstream path **614** from the master communication device **102** to the service communication device **106A** in the LBMN **600B** may be via the optical routing devices **104A**, **104B**, **104D**, and **104E**. In this case, in the downstream path **612**, the optical routing device **104B**, which is a passive optical node, may include a free-space optical beam splitter **618** (e.g., a plate beam splitter) that splits the received laser beam in the downstream path **612** into two (or more) separate laser beams (as shown), where one laser beam from such optical split is directed towards the optical routing device **104D** (which may be a passive optical routing device) and the other laser beam from such optical split is directed towards the optical routing device **104C** to form the LBMN **600B**. In an implementation, such split by the free-space optical beam splitter **618** may cause the received laser beam in the downstream path **612** in a first wavelength to split into two separate laser beams having same wavelengths (i.e., the first wavelength). In another implementation, such optical split by the free-space optical beam splitter **618** may cause the received laser beam in the downstream path **612** in a first wavelength to split into two separate laser beams having different wavelengths. Further, in this case, in the upstream path **614**, the optical routing device **104B** may further include a free-space optical beam combiner **620** (e.g., a plate beam combiner) that combines two received laser beams (one each from the optical routing devices **104D** and **104C**) in the upstream path **614** into a single laser beam (as shown) to form the LBMN **600B** for upstream data communication. Similarly, another laser beam-based communication route in the downstream path **612** and the upstream path **614** from the master communication device **102** to the service communication device **106B** in the LBMN **600B** may be via the optical routing devices **104A**, **104B**, and **104C**. The processor **306** may be further configured to receive an instruction via the one or more RF supervisory links to control a movement of the mounting component **302** along with the optical routing component **304** such that an angle or a direction of deflection of one or more laser beams from the optical routing component **304** is changed. In this case, the optical routing component **304** may comprise the one or more laser beam handling regions **318** configured to handle a plurality of laser beams concurrently in which the first laser beam **604** may be deflected via the first region **318A** of the one or more laser beam handling regions **318** for downstream data communication in the downstream path **612** and a second laser beam **606** is deflected via the second region **318B** of the one or more laser beam handling regions **318** for upstream data communication in the upstream path **614**. Alternatively, the optical routing component **304** may be configured to handle one laser beam over-the-air in the laser-beam based wireless communication network for both downstream and upstream data communication with the master communication device **102** and the one or more service communication devices **106**, such as the service communication devices **106A** and **106B**. In other words, there are two ways of communication to form the laser-beam based wireless backhaul (i.e., the laser-beam based wireless communication network). In a first implementation, there may be two laser beams (the first laser beam **604** and second laser beam **606**) for a free-space laser communication with separate transmission (Tx) and reception (Rx). In a second implementation, called bi-directional, there may be single laser beam for Tx and Rx but at two different wavelengths using WDM. The wavelength splitting and combining may be performed passively using a dichroic mirror. Thus, a full coverage with high signal-to-noise (SNR) in an indoor area with high-speed data connectivity to end-user devices, such as the UE **108A** and the UE **108B** may be provided without the need to deploy conventional RF intermediate routers simplifying the deployment while significantly improving the data throughput as compared to conventional systems.

[0113] FIGS. 7A, 7B, and 7C, collectively, is a flowchart that illustrates an exemplary method for operating an optical routing device for ultra-flexible and ultra-reliable laser beam based wireless communication for indoor use, in accordance with an embodiment of the disclosure. FIGS. 7A, 7B,

and 7C are explained in conjunction with elements from FIGS. 1, 2, 3A to 3C, 4, 5, 6A, and 6B. With reference to FIGS. 7A, 7B, and 7C there is shown a flowchart **700** comprising exemplary operations **702** through **734**. The operations **702** to **734** may be executed by the optical routing device **104A**.

[0114] At **702**, a radio frequency (RF) supervisory link with the master communication device **102** or one or more RF supervisory links with one or more other optical routing devices of the plurality of optical routing devices **104** and/or one or more service communication devices **106**, may be established. In a case where the optical routing device **104A** is a primary optical deflector node, the processor **306** may be configured to establish the RF supervisory link with the master communication device **102**. In a case where the optical routing device **104A** is a secondary optical deflector node, the processor **306** may be configured to establish the one or more RF supervisory links with one or more other optical routing devices of the plurality of optical routing devices **104** and/or the one or more service communication devices **106**.

[0115] At **704**, communication (e.g., of control signals or sensor data etc.) may be made over-the-air with the master communication device **102** or the one or more service communication devices **106** via the one or more radio frequency (RF) supervisory links.

[0116] At **706**, an instruction may be received via the one or more RF supervisory links to control a movement of the mounting component **302** along with the optical routing component **304** such that an angle or a direction of deflection (or routing after beam splitting) of one or more laser beams from the optical routing component **304** is changed. The operation **706** may include one or more sub-operations **706A** to **706E**.

[0117] At **706A**, a pitch rotation of the mounting component **302** may be controlled along an x-axis **330A** that is parallel to a mounting surface **328** of the optical routing device **104A**. The control of the movement may be in response to the instruction received from the master communication device **102** or the cloud server **116** via the one or more RF supervisory links. At **706B**, a yaw rotation of the mounting component **302** may be controlled along the y-axis **330B** orthogonal to the x-axis **330A**. At **706C**, a roll rotation of the mounting component **302** may be controlled along the z-axis **330C**. At **706D**, an alignment of an optical path may be caused among the master communication device **102**, one or more other optical routing devices, and/or the one or more service communication devices **106**, based on the controlling of one or more of the pitch rotation, the yaw rotation, or the roll rotation of the mounting component **302**. At **706E**, a transmit polarization of the one or more laser beams (e.g., the first laser beam **326A** or **604** and the second laser beam **326B** or **606**) deflected by the optical routing component **304** may be aligned in accordance with a destination device when an optical polarization multiplexing is utilized in data communication over a laser beam-based wireless communication network. The destination device may be one of the master communication device **102**, one or more other optical routing devices, or the one or more service communication devices **106**. The control may then pass to **708A** or **798B** depending on a use case.

[0118] At **708A**, a plurality of laser beams may be concurrently handled (i.e., deflected or focussed) via the one or more laser beam handling regions **318** in which the first laser beam (e.g., the first laser beam **326A** or **604**) may be deflected via the first region **318A** of the one or more laser beam handling regions **318** for downstream data communication in a downstream path and a second laser beam (e.g., the second laser beam **326B** or **606**) may be deflected via the second region **318B** of the one or more laser beam handling regions **318** for upstream data communication in an upstream path. In accordance with an embodiment, different wavelengths may be utilized for the first laser beam (e.g., at 1550 nm band) and the second laser beam (e.g., at 1300 nm band) for downstream and upstream data communication respectively.

[0119] At **708B**, a laser beam may be handled (i.e., deflected, focused, or routed) by one or more laser beam handling regions of the optical routing component **304** over-the-air in a laser-beam based wireless communication network for both downstream and upstream data communication

with the master communication device **102** and the one or more service communication devices **106**. In this implementation, the laser beam may comprise a plurality of data streams modulated in at least one of a coarse wavelength division multiplexing (CWDM), a dense wavelength division multiplexing (DWDM), or another wavelength division multiplexing, for the downstream and upstream data communication. In this case, a single laser beam includes both Tx and Rx communication paths but at two different wavelengths done using WDM. The wavelength splitting and combining may be performed passively using a dichroic mirror. Additional wavelength channels (i.e., communication data lanes) may offer a path to increasing the data rate. For example, coarse wavelength division multiplexing (CWDM) may allow up to 18 channels and dense wavelength division multiplexing (DWDM) may allow up to 200 channels.

[0120] At **710**, a Laser Beam Mesh Network (LBMN) or a Laser Beam Cascaded Network (LBCN) may be generated in coordination with the master communication device **102**, in which the optical routing device is a part of the LBMN and the LBCN.

[0121] At **712**, an Optical Wireless Charging-Cum-Networking (OWCCN) mode may be selected and activated in which at least one of the one or more laser beam handling regions **318** may be configured to deflect a laser beam to form a laser beam-based wireless communication network and concomitantly cause the same laser beam to charge the rechargeable battery **314** via the at least one of the one or more laser beam handling regions **318**. The processor **306** may be further configured to activate the OWCCN mode. In an example, the decision of which deflection region to select from the one or more laser beam handling regions **318** to charge the rechargeable battery **314** may be based on the data traffic in the upstream path or the downstream path. For example, the second region **318B** of the one or more laser beam handling regions **318** may be selected if data traffic in the downstream path is less than the upstream path if the second region **318B** is used for the downstream data communication in the downstream path. In another example, the decision of which deflection region to select from the one or more laser beam handling regions **318** may not be based on data traffic and may be predetermined. Alternatively, one of the one or more laser beam handling regions, such as the second region **318B**, is configured to handle an additional laser beam (e.g., power laser or beacon laser beam at around 800 nm) over-the-air for an optical wireless power transmission to remotely charge a corresponding rechargeable battery of each of one or more other optical routing devices and/or the service communication device **106A**.

[0122] At **714**, a connection from the selected deflection region of the one or more laser beam handling regions **318** to the rechargeable battery **314**, may be intermittently enabled and disabled. The processor **306** may be further configured to intermittently enable and disable the connection from the selected deflection region to the rechargeable battery **314** of the optical routing device **104A**.

[0123] At **716**, the connection to the rechargeable battery **314** may be switched from among the one or more laser beam handling regions **318**. In other words, the connection of the first region **318A** to the rechargeable battery **314** via a first photovoltaic cell of the charging handler **316B** may be periodically disabled, whereas the connection of the second region **318B** to the rechargeable battery **314** via a second photovoltaic cell of the charging handler **316B** may be periodically enabled, and vice-versa.

[0124] At **718**, an orientation and a level of the optical routing component **304** may be measured with respect to the reference surface **334** by use of the one or more sensors **322** to maintain an alignment of an optical path with one or more of the master communication device **102**, one or more other optical routing devices, and/or the one or more service communication devices **106**.

[0125] At **720**, the measured orientation, the level, and location data of the optical routing device **104A** may be communicated over the one or more RF supervisory links to the master communication device **102** or the cloud server **116**. The processor **306** may be further configured to communicate the measured orientation, the level, and location data of the optical routing device **104A** to the master communication device **102** or the cloud server **116**.

[0126] At **722**, performance data of the laser beam-based wireless communication network (i.e., the second communication network **120**) may be periodically communicated to the master communication device **102** or the cloud server **116**.

[0127] At **724**, a thermomechanical-induced change may be determined in an alignment of an optical path of the optical routing device **104A** with one or more of: the master communication device **102**, one or more other optical routing devices, and/or the one or more service communication devices **106**. The processor **306** may be further configured to determine the thermomechanical-induced change.

[0128] At **726**, an automatic alignment of the optical path may be executed when the temperature monitored by the temperature sensor **336** around the optical routing device **104A** in the defined indoor area **112** is outside a pre-set range. The processor **306** may be further configured to execute the automatic alignment of the optical path.

[0129] At **728**, a Laser Beam Network Control (LBNC) instructions may be received from the master communication device **102** or the cloud server **116**. The processor **306** may be further configured to receive the LBNC instructions from the master communication device **102** or the cloud server **116** over the one or more RF supervisory links.

[0130] At **730**, an orientation of the one or more laser beam handling regions **318** in the deflecting surface **320** of the optical routing component **304** may be dynamically adjusted based on the LBNC instructions received from the master communication device **102** or the cloud server **116**. The instruction to control the movement of the mounting component **302** may be received from the master communication device **102** or the cloud server **116** via the one or more RF supervisory links. The processor **306** may be further configured to switch and re-align the one or more deflected laser beams **326A1** and **326B1** in a real-time or a near real-time to a different destination device based on a change in the angle or the direction of deflection of the one or more laser beams from the optical routing component **304**.

[0131] At **732**, a polarization of the plurality of laser beams, such as the first laser beam **326A** and the second laser beam **326B**, may be controlled via the one or more polarizers **342A** and **342B** of the optical routing device **104A**. In an implementation, the one or more beam-splitters **340A** and **340B** may be extended or retracted from the mounting component **302** based on the LBNC instructions received from the master communication device **102** or the cloud server **116**.

[0132] At **734**, one of the optical path options **312C** that may be different fallback laser-link options may be selected when a current optical path option is interrupted. The optical path options **312C** may be different fall-back laser-link options available to a given optical routing device, such as the optical routing device **104A**, to establish a laser beam connectivity with at least one of: the master communication device **102**, one or more other optical routing devices, and/or the one or more service communication devices **106**.

[0133] Various embodiments of the disclosure may provide the optical routing device **104A** (FIGS. **1**, **3A**, **3B**, and **3C**). The optical routing device **104A** may comprise the mounting component **302** and the optical routing component **304** attached to the mounting component **302**. The mounting component **302** may comprise the rechargeable battery **314** to power operations of the optical routing device **104A**. The mounting component **302** may further comprise the processor **306** configured to communicate over-the-air with the master communication device **102** or one or more service communication devices **106** via one or more radio frequency (RF) supervisory links. The processor **306** may be further configured to receive an instruction via the one or more RF supervisory links to control a movement of the mounting component **302** along with the optical routing component **304** such that an angle or a direction of deflection of one or more laser beams from the optical routing component **304** is changed. The optical routing component **304** may comprise the one or more laser beam handling regions **318** configured to handle a plurality of laser beams concurrently in which the first laser beam **326A** may be deflected via a first region **318A** of the one or more laser beam handling regions **318** for downstream data communication in a

downstream path and a second laser beam **326B** is deflected via a second region **318B** of the one or more laser beam handling regions **318** for upstream data communication in an upstream path. [0134] Various embodiments of the disclosure may further provide the optical routing device **104A** (FIGS. **1**, **3A**, **3B**, and **3C**). The optical routing device **104A** may comprise the mounting component **302** and the optical routing component **304** attached to the mounting component **302**. The mounting component **302** may comprise the rechargeable battery **314** to power operations of the optical routing device **104A**. The mounting component **302** may further comprise the processor **306** configured to communicate over-the-air with the master communication device **102** or one or more service communication devices **106** via one or more radio frequency (RF) supervisory links. The processor **306** may be further configured to receive an instruction via the one or more RF supervisory links to control a movement of the mounting component **302** along with the optical routing component **304** such that an angle or a direction of deflection of one or more laser beams from the optical routing component **304** is changed. The optical routing component **304** may comprise the one or more laser beam handling regions **318** configured to handle a laser beam over-the-air in a laser-beam based wireless communication network for both downstream and upstream data communication with the master communication device **102** and the one or more service communication devices **106**.

[0135] In accordance with an embodiment, wherein the laser beam may comprise a plurality of data streams modulated in at least one of: a coarse wavelength division multiplexing (CWDM), a dense wavelength division multiplexing (DWDM), or another wavelength division multiplexing, for the downstream and upstream data communication.

[0136] While various embodiments described in the present disclosure have been described above, it should be understood that they have been presented by way of example, and not limitation. It is to be understood that various changes in form and detail can be made therein without departing from the scope of the present disclosure. In addition to using hardware (e.g., within or coupled to a central processing unit (“CPU”), microprocessor, micro controller, digital signal processor, processor core, system on chip (“SOC”) or any other device), implementations may also be embodied in software (e.g. computer readable code, program code, and/or instructions disposed in any form, such as source, object or machine language) disposed for example in a non-transitory computer-readable medium configured to store the software. Such software can enable, for example, the function, fabrication, modeling, simulation, description and/or testing of the apparatus and methods describe herein. For example, this can be accomplished through the use of general program languages (e.g., C, C++), hardware description languages (HDL) including Verilog HDL, VHDL, and so on, or other available programs. Such software can be disposed in any known non-transitory computer-readable medium, such as semiconductor, magnetic disc, or optical disc (e.g., CD-ROM, DVD-ROM, etc.). The software can also be disposed as computer data embodied in a non-transitory computer-readable transmission medium (e.g., solid state memory any other non-transitory medium including digital, optical, analog-based medium, such as removable storage media). Embodiments of the present disclosure may include methods of providing the apparatus described herein by providing software describing the apparatus and subsequently transmitting the software as a computer data signal over a communication network including the internet and intranets.

[0137] It is to be further understood that the system described herein may be included in a semiconductor intellectual property core, such as a microprocessor core (e.g., embodied in HDL) and transformed to hardware in the production of integrated circuits. Additionally, the system described herein may be embodied as a combination of hardware and software. Thus, the present disclosure should not be limited by any of the above-described exemplary embodiments but should be defined only in accordance with the following claims and their equivalents.

Claims

1. A first optical routing device, comprising: a mounting component; a memory configured to store optical path options, wherein the optical path options are different fall-back laser-link options available to the first optical routing device to establish a laser beam connectivity; and an optical routing component attached to the mounting component, wherein the mounting component comprises: a rechargeable battery configured to power operations of the first optical routing device; and a processor configured to: communicate over-the-air with one of a master communication device or one or more service communication devices via one or more radio frequency (RF) supervisory links; receive an instruction via the one or more RF supervisory links to control a movement of the mounting component along with the optical routing component; and perform a range measurement to compute a distance between the first optical routing device and an optical node, wherein the different fall-back laser-link options for the first optical routing device comprises location coordinates of the first optical routing device and corresponding two or more neighbouring optical routing devices and laser beam alignment information, and the laser beam alignment information comprises an angle of deflection of a laser beam of one or more laser beams, a direction of deflection of the laser beam, a beam width of the laser beam of the one or more laser beams, and a laser intensity for the laser beam between two optical routing devices among the first optical routing device and the corresponding two or more neighbouring optical routing devices.
2. The first optical routing device according to claim 1, wherein the optical routing component is configured to deflect the one or more laser beams over-the-air for free-space optical communication among the master communication device and the one or more service communication devices via at least the first optical routing device, and the deflection of the one or more laser beams is independent of optical fibers.
3. The first optical routing device according to claim 2, wherein the first optical routing device is a passive optical node that is independent of amplification of signals in the deflected one or more laser beams, and each of the master communication device and the one or more service communication devices is one of a wireless access point, a wireless router, a home gateway device, a fixed wireless access (FWA) device, or a network controller.
4. The first optical routing device according to claim 1, wherein the processor is further configured to adjust, based on the range measurement, beam divergence of the one or more laser beams for data signal transmission across a laser beam based wireless network.
5. The first optical routing device according to claim 1, wherein the processor is further configured to control generation of one of a Laser Beam Mesh Network (LBMN) or a Laser Beam Cascaded Network (LBCN) in coordination with the master communication device, and the first optical routing device is a part of the generated one of the LBMN or the LBCN.
6. The first optical routing device according to claim 1, wherein the optical routing component includes one or more laser beam handling regions, and the processor is further configured to activate an Optical Wireless Charging-Cum-Networking (OWCCN) mode in which at least one of the one or more laser beam handling regions is configured to deflect a laser beam of the one or more laser beams to establish a laser beam-based wireless communication network, and concomitantly cause the laser beam to charge the rechargeable battery via the at least one of the one or more laser beam handling regions.
7. The first optical routing device according to claim 6, wherein the processor is further configured to intermittently enable and disable a connection of the at least one of the one or more laser beam handling regions to the rechargeable battery.
8. The first optical routing device according to claim 6, wherein the processor is further configured to switch a connection of the rechargeable battery among the one or more laser beam handling regions based on a Status-of-charge (SoC) of the rechargeable battery that is below a defined SoC.

- 9.** The first optical routing device according to claim 1, further comprising a temperature sensor, wherein the processor is further configured to determine a thermomechanical-induced change in an alignment of an optical path of the first optical routing device with one or more of the master communication device, one or more second optical routing devices, and the one or more service communication devices.
- 10.** The first optical routing device according to claim 9, wherein the temperature sensor is configured to monitor a temperature, the temperature is outside a defined range, and the processor is further configured to execute an automatic alignment of the optical path based on the temperature that is outside the defined range.
- 11.** The first optical routing device according to claim 1, wherein the processor is further configured to receive Laser Beam Network Control (LBNC) instructions from one of the master communication device or a cloud server.
- 12.** The first optical routing device according to claim 11, wherein the processor is further configured to dynamically adjust an orientation of the optical routing component, based on the LBNC instructions received from the one of the master communication device or the cloud server.
- 13.** The first optical routing device according to claim 1, wherein the instruction to control the movement of the mounting component is received from one of the master communication device or a cloud server via the one or more RF supervisory links, and the processor is further configured to switch and re-align the one or more laser beams in one of a real-time or a near real-time to a different destination device based on a change in the one of the angle or the direction of deflection of the one or more laser beams from the optical routing component.
- 14.** The first optical routing device according to claim 1, wherein the optical routing component further comprises one or more beam-splitters, each beam-splitter of the one or more beam-splitters is configured to one of split a laser beam of a plurality of laser beams into two or more laser beams or combine multiple laser beams of the plurality of laser beams into a single beam, and the plurality of laser beams includes the one or more laser beams.
- 15.** The first optical routing device according to claim 14, wherein the one or more beam-splitters are movably mountable on the mounting component, and the processor is further configured to: receive Laser Beam Network Control (LBNC) instructions from one of the master communication device or a cloud server; and one of extend or retract the one or more beam-splitters from the mounting component based on the LBNC instructions received from one of the master communication device or the cloud server.
- 16.** The first optical routing device according to claim 1, wherein the optical routing component further comprises a polarizer, and the processor is further configured to control a polarization of the one or more laser beams via the polarizer.
- 17.** The first optical routing device according to claim 1, wherein the instruction corresponds to a change in one of an angle or a direction of deflection of one or more laser beams from the optical routing component.
- 18.** The first optical routing device according to claim 1, wherein the optical node is one of a second optical routing device, the master communication device, or one of the one or more service communication devices.
- 19.** A first optical routing device, comprising: a mounting component; a memory configured to store optical path options, wherein the optical path options are different fall-back laser-link options available to the first optical routing device to establish a laser beam connectivity; and an optical routing component attached to the mounting component, wherein the mounting component comprises: a rechargeable battery configured to power operations of the first optical routing device; and a processor configured to: communicate over-the-air with one of a master communication device or one or more service communication devices via one or more radio frequency (RF) supervisory links; receive an instruction via the one or more RF supervisory links to control a movement of the mounting component along with the optical routing component; and establish an

RF supervisory link with the master communication device, in a case where the first optical routing device is a primary optical deflector node, wherein the primary optical deflector node is the first optical routing device that is configured to send and receive a laser beam of the one or more laser beams from the master communication device, the first optical routing device is in one of a line-of-sight of the master communication device or a nearest optical node to the master communication device, the different fall-back laser-link options for the first optical routing device comprises location coordinates of the first optical routing device and corresponding two or more neighbouring optical routing devices and laser beam alignment information, and the laser beam alignment information comprises an angle of deflection of a laser beam of the one or more laser beams, a direction of deflection of the laser beam, a beam width of the laser beam of the one or more laser beams, and a laser intensity for the laser beam between two optical routing devices among the first optical routing device and the corresponding two or more neighbouring optical routing devices.

20. The first optical routing device according to claim 19, wherein the processor is further configured to establish the one or more RF supervisory links with one of one or more second optical routing devices or the one or more service communication devices, in a case where the first optical routing device is a secondary optical deflector node, and the secondary optical deflector node is in one of a non-line-of-sight (NLOS) to the master communication device or not directly able to send and receive a laser beam of the one or more laser beams from the master communication device.
